

GraphTheorySymbol: L^AT_EX Commands with Symbols for Graph Theory

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Abstract

This documentation is for L^AT_EX package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

Contents

1	Introduction	3
2	Usage	4
2.1	Memo or for the impatient	4
2.2	Detailed usage	7
2.2.1	General macros	7
2.2.2	Standard adjacency macros	7
2.2.3	Standard incidence macros	11
2.2.4	Monochrome adjacency macros	15
2.2.5	Monochrome incidence macros	19
2.2.6	Meta-logic macros	23
2.3	Good practices	25
3	Implementation	25
4	Acknowledgments	39
5	FAQ	39
6	Support	40
7	Change History	40

1 Introduction

This documentation is for $\text{\LaTeX}$ package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

This package is licenced under Lesser GNU General Public License version 3 or later (LGPLv3+). It means you are free do anything with it, but if you do improve the code of the symbols that we provide, please do share these improvements in a similar way. There is no restriction on other graph theory symbols/code, even if it would be a good fit for inclusion in this package. There is also no need to share in a similar way what you build with this package. The text of the licenses is available at:

<https://www.gnu.org/licenses/>.

The git repository of this source code is also available at:

<https://github.com/LLyaudet/GraphTheorySymbol/>.

Clone it with this URL:

[git@github.com:LLyaudet/GraphTheorySymbol.git](https://github.com/LLyaudet/GraphTheorySymbol.git).

This package is available on CTAN:

<https://ctan.org/pkg/graph-theory-symbol>.

This is our first $\text{\LaTeX}$ package. It may be that our macros are quite slow and that a document making extensive use of these macros will be slow to compile. If you know how to produce better $\text{\LaTeX}$ code, please submit a pull request on GitHub.

2 Usage

2.1 Memo or for the impatient

Add in your preamble: `\usepackage{graph-theory-symbol}`.

For clarity, below we only show long macro names and short macro names; see Detailed usage section for intermediate length macro names.

Long name	Short name	Result	In context
<code>\GTSadjacency/</code>	<code>\GTSa/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyRelation/</code>	<code>\GTSaR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacency/</code>	<code>\GTSu/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyRelation/</code>	<code>\GTSuR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUp/</code>	<code>\GTSdAU/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpRelation/</code>	<code>\GTSdAUR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUp/</code>	<code>\GTSnDAU/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpRelation/</code>	<code>\GTSnDAUR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDown/</code>	<code>\GTSdAD/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownRelation/</code>	<code>\GTSdADR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDown/</code>	<code>\GTSnDAD/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownRelation/</code>	<code>\GTSnDADR/</code>	$\circ$	$u \circ v$
<code>\GTSincidence/</code>	<code>\GTSi/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceRelation/</code>	<code>\GTSiR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidence/</code>	<code>\GTSnI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceRelation/</code>	<code>\GTSnIR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidence/</code>	<code>\GTSoI/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceRelation/</code>	<code>\GTSoIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidence/</code>	<code>\GTSnOI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceRelation/</code>	<code>\GTSnOIR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidence/</code>	<code>\GTSiI/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceRelation/</code>	<code>\GTSiIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidence/</code>	<code>\GTSnII/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceRelation/</code>	<code>\GTSnIIR/</code>	$\circ$	$v \circ e$
Long name	Short name	Result	In context
<code>\GTSadjacencyMonochrome/</code>	<code>\GTSaM/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyMonochromeRelation/</code>	<code>\GTSaMR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochrome/</code>	<code>\GTSuM/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochromeRelation/</code>	<code>\GTSuMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochrome/</code>	<code>\GTSdAUM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSdAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochrome/</code>	<code>\GTSnDAUM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSnDAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochrome/</code>	<code>\GTSdADM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSdADM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochrome/</code>	<code>\GTSnDADM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSnDADM/</code>	$\circ$	$u \circ v$
<code>\GTSincidenceMonochrome/</code>	<code>\GTSiM/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceMonochromeRelation/</code>	<code>\GTSiMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochrome/</code>	<code>\GTSnIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochromeRelation/</code>	<code>\GTSnIMR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochrome/</code>	<code>\GTSoIM/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochromeRelation/</code>	<code>\GTSoIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochrome/</code>	<code>\GTSnOIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochromeRelation/</code>	<code>\GTSnOIMR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochrome/</code>	<code>\GTSiIM/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochromeRelation/</code>	<code>\GTSiIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochrome/</code>	<code>\GTSnIIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochromeRelation/</code>	<code>\GTSnIIMR/</code>	$\circ$	$v \circ e$

\GTSsetMainColor{blue}			
Long name	Short name	Result	In context
\GTSadjacency/	\GTSa/		$u \overset{\circ}{\circ} v$
\GTSadjacencyRelation/	\GTSaR/		$u \overset{\circ}{\circ} v$
\GTSunadjacency/	\GTSu/		$u \overset{\circ}{\circ} v$
\GTSunadjacencyRelation/	\GTSuR/		$u \overset{\circ}{\circ} v$
\GTSdirectedAdjacencyUp/	\GTSdAU/		$u \overset{\circ}{\circ} v$
\GTSdirectedAdjacencyUpRelation/	\GTSdAUR/		$u \overset{\circ}{\circ} v$
\GTSnegatedDirectedAdjacencyUp/	\GTSnDAU/		$u \overset{\circ}{\circ} v$
\GTSnegatedDirectedAdjacencyUpRelation/	\GTSnDAUR/		$u \overset{\circ}{\circ} v$
\GTSdirectedAdjacencyDown/	\GTSdAD/		$u \overset{\circ}{\circ} v$
\GTSdirectedAdjacencyDownRelation/	\GTSdADR/		$u \overset{\circ}{\circ} v$
\GTSnegatedDirectedAdjacencyDown/	\GTSnDAD/		$u \overset{\circ}{\circ} v$
\GTSnegatedDirectedAdjacencyDownRelation/	\GTSnDADR/		$u \overset{\circ}{\circ} v$
\GTSincidence/	\GTSi/		$v \overset{\circ}{\circ} e$
\GTSincidenceRelation/	\GTSiR/		$v \overset{\circ}{\circ} e$
\GTSnegatedIncidence/	\GTSnI/		$v \overset{\circ}{\circ} e$
\GTSnegatedIncidenceRelation/	\GTSnIR/		$v \overset{\circ}{\circ} e$
\GTSoutIncidence/	\GTSOI/		$v \overset{\circ}{\circ} e$
\GTSoutIncidenceRelation/	\GTSOIR/		$v \overset{\circ}{\circ} e$
\GTSnegatedOutIncidence/	\GTSnOI/		$v \overset{\circ}{\circ} e$
\GTSnegatedOutIncidenceRelation/	\GTSnOIR/		$v \overset{\circ}{\circ} e$
\GTSinIncidence/	\GTSiI/		$v \overset{\circ}{\circ} e$
\GTSinIncidenceRelation/	\GTSiIR/		$v \overset{\circ}{\circ} e$
\GTSnegatedInIncidence/	\GTSnII/		$v \overset{\circ}{\circ} e$
\GTSnegatedInIncidenceRelation/	\GTSnIIR/		$v \overset{\circ}{\circ} e$

Long name	Short name	Result	In context
\GTSmetaNot/	\GTSmN/	$\neg$	$\neg \overset{\circ}{\circ}$
\GTSmetaAnd/	\GTSmA/	$\wedge$	$\overset{\circ}{\circ} \wedge \overset{\circ}{\circ}$
\GTSmetaOr/	\GTSmO/	$\vee$	$\overset{\circ}{\circ} \vee \overset{\circ}{\circ}$

Meta-logic can be used for compound adjacency types, etc.

Implicit meta-and via concatenation, no space except around whole relations compositions, no parentheses via higher priority to meta-and, brackets when all possible adjacencies are specified between two vertices:

Tournament adjacency:

$\forall u, v, u \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} v$.

Tournament adjacency with maybe some colored edge also, who knows?:

$\forall u, v, u \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} v$.

Directed graph:

$\forall u, v, u \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} v$.

Oriented graph:

$\forall u, v, u \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} v$.

Tournament adjacency:

$\overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ}$ is obtained with $\backslash\mathrm{mathrel}\{\backslash\mathrm{GTSdAD}/\backslash\mathrm{GTSnDAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSdAU}/\}$.

Tournament adjacency with maybe some colored edge also, who knows?:

$\overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ}$ is obtained with $\backslash\mathrm{mathrel}\{\backslash\mathrm{GTSdAD}/\backslash\mathrm{GTSnDAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSdAU}/\}$.

Directed graph:

$\overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ}$ is obtained with

$\backslash\mathrm{mathrel}\{\backslash\mathrm{GTSdAD}/\backslash\mathrm{GTSnDAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSdAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSdAD}/\backslash\mathrm{GTSdAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSnDAU}/\}$.

Oriented graph:

$\overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ}$ is obtained with

$\backslash\mathrm{mathrel}\{\backslash\mathrm{GTSdAD}/\backslash\mathrm{GTSnDAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSdAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSnDAU}/\}$.

Prefix notation:

$\exists u, v, u \left[\overset{m}{V} \overset{m}{\overset{m}{\overset{m}{\circ}}} \right] v \Leftrightarrow \exists u, v, u \left[\overset{m}{\overset{m}{\overset{m}{\circ}}} \overset{m}{V} \overset{m}{\overset{m}{\circ}} \right] v.$

$\exists v, e, v \left[\overset{m}{V} \overset{m}{\overset{m}{\overset{m}{\circ}}} \right] e \Leftrightarrow \exists v, e, v \left[\overset{m}{\overset{m}{\overset{m}{\circ}}} \overset{m}{V} \overset{m}{\overset{m}{\circ}} \right] e.$

$\left[\overset{m}{V} \overset{m}{\overset{m}{\overset{m}{\circ}}} \right]$ is obtained with

`\mathrel{\{[\backslashGTSmO/\backslashGTSxa[mC=red,d]\backslashGTSxa[mC=green,d]\backslashGTSxa[mC=blue,u]]\}}.`

$\left[\overset{m}{\overset{m}{\overset{m}{\circ}}} \right]$ is obtained with

`\mathrel{\{[\backslashGTSmO/\backslashGTSxi[mC=red,i]\backslashGTSxi[mC=green,i]\backslashGTSxi[mC=blue,o]]\}}.`

Apart from these symbol macros, versatile macros to call them all are available with options:

`\GTSxa[]`, `\GTSxadj[]`, or `\GTSxadjacency[]` for adjacency symbols,

`\GTSxi[]`, `\GTSxinc[]`, or `\GTSxincidence[]` for incidence symbols,

`\GTSxml[]`, or `\GTSxmetallogic[]` for meta-logic symbols,

`\GTSx[]{} for all symbols`

`(\GTSx[]{}a} = \GTSxa[], \GTSx[]{}i} = \GTSxi[], \GTSx[]{}m} = \GTSxml[]).`

See Detailed usage section for full syntax of these macros.

2.2 Detailed usage

First you need to add in your preamble: `\usepackage{graph-theory-symbol}`.

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

For completeness, we also documented macros with an at sign, but as usual, you should not (have to) use them directly.

2.2.1 General macros

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{<commandname>}` The name of the command to create.

`{<commandcode>}` The code of the command to create.

`\GTS@newprotectedcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{<commandname>}` The name of the command to create.

`{<commandcode>}` The code of the command to create.

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

`{<color>}` The color that must be used by GraphTheorySymbol afterward.

`\GTSsetSecondColor` This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

`{<color>}` The color that must be used by GraphTheorySymbol afterward.

2.2.2 Standard adjacency macros

`\GTS@a` This macro will draw any of the adjacency symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and `\GTS@b`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

`{<mainColor>}` If non-empty, this color will be used as main color.

`{<secondColor>}` If non-empty, this color will be used as second color.

`{<drawSlash>}` Non-empty implies that a negation will be drawn.

`{<drawUpArrow>}` Non-empty implies that an up-arrow will be drawn.

`{<drawDownArrow>}` Non-empty implies that a down-arrow will be drawn.

`\GTS@b` This macro builds on top of `\GTS@a` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@a`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

`{<mainColor>}` If non-empty, this color will be used as main color.

`{<secondColor>}` If non-empty, this color will be used as second color.

`{<drawSlash>}` Non-empty implies that a negation will be drawn.

`{<drawUpArrow>}` Non-empty implies that an up-arrow will be drawn.

`{<drawDownArrow>}` Non-empty implies that a down-arrow will be drawn.

`{<relation>}` Non-empty implies that relation spacing will be added.

\GTSxadjacency, \GTSxadj, \GTSxa Versatile macros for adjacency that can be given options.

Argument:

[*(optionsList)*] Sets the following options if not empty:

mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
 secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
 monochrome or mono or mon or m (equivalent to mC=black,sC=black),
 negated or not or n,
 up or u,
 down or d,
 relation or rel or r.

\GTSadjacency, \GTSadj, \GTSa This macro will draw an adjacency symbol. A first drawing didn't include the small dots in the middle of the two circles corresponding to the graph vertices. But it was too similar to “spoon” and multimap symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property to remind of the classical textual notation of adjacency relation as Adj(..).

`A.\GTSadjacency/A.`

or `A.\GTSadj/A.`

or `A.\GTSa/A.`

will yield $A \circ A.$

`u \GTSadjacency/ v`

or `u \GTSadj/ v`

or `u \GTSa/ v`

will yield $u \circ v.$

`$u \GTSadjacency/ v$`

or `$u \GTSadj/ v$`

or `$u \GTSa/ v$`

will yield $u \circ v.$

\GTSadjacencyRelation, \GTSadjRel, \GTSaR This macro will draw an adjacency symbol.

`$u\GTSadjacencyRelation/v$`

or `$u\GTSadjRel/v$`

or `$u\GTSaR/v$`

will yield $u \circ v.$

\GTSunadjacency, \GTSunadj, \GTSu This macro will draw an unadjacency symbol.

`A.\GTSunadjacency/A.`

or `A.\GTSunadj/A.`

or `A.\GTSu/A.`

will yield $A \not\circ A.$

`u \GTSunadjacency/ v`

or `u \GTSunadj/ v`

or `u \GTSu/ v`

will yield $u \not\circ v.$

`$u \GTSunadjacency/ v$`

or $\$u \backslash \text{GTSunadj} / v\$$
or $\$u \backslash \text{GTSu} / v\$$
will yield $u \int v$.

$\backslash \text{GTSunadjacencyRelation}$, $\backslash \text{GTSunadjRel}$, $\backslash \text{GTSuR}$ This macro will draw an unadjacency symbol.

$\$u \backslash \text{GTSunadjacencyRelation} / v\$$
or $\$u \backslash \text{GTSunadjRel} / v\$$
or $\$u \backslash \text{GTSuR} / v\$$
will yield $u \int v$.

$\backslash \text{GTSdirectedAdjacencyUp}$, $\backslash \text{GTSdirAdjUp}$, $\backslash \text{GTSdAU}$ This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

$A. \backslash \text{GTSdirectedAdjacencyUp} / A.$
or $A. \backslash \text{GTSdirAdjUp} / A.$
or $A. \backslash \text{GTSdAU} / A.$
will yield $A. \int A.$

$u \backslash \text{GTSdirectedAdjacencyUp} / v$
or $u \backslash \text{GTSdirAdjUp} / v$
or $u \backslash \text{GTSdAU} / v$
will yield $u \int v$.

$\$u \backslash \text{GTSdirectedAdjacencyUp} / v\$$
or $\$u \backslash \text{GTSdirAdjUp} / v\$$
or $\$u \backslash \text{GTSdAU} / v\$$
will yield $u \int v$.

$\backslash \text{GTSdirectedAdjacencyUpRelation}$, $\backslash \text{GTSdirAdjUpRel}$, $\backslash \text{GTSdAUR}$ This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

$\$u \backslash \text{GTSdirectedAdjacencyUpRelation} / v\$$
or $\$u \backslash \text{GTSdirAdjUpRel} / v\$$
or $\$u \backslash \text{GTSdAUR} / v\$$
will yield $u \int v$.

$\backslash \text{GTSnegatedDirectedAdjacencyUp}$, $\backslash \text{GTSnegDirAdjUp}$, $\backslash \text{GTSnDAU}$ This macro will draw a negated directed adjacency upward symbol.

$A. \backslash \text{GTSnegatedDirectedAdjacencyUp} / A.$
or $A. \backslash \text{GTSnegDirAdjUp} / A.$
or $A. \backslash \text{GTSnDAU} / A.$
will yield $A. \int A.$

$u \backslash \text{GTSnegatedDirectedAdjacencyUp} / v$
or $u \backslash \text{GTSnegDirAdjUp} / v$
or $u \backslash \text{GTSnDAU} / v$

will yield $u \nrightarrow v$.

`$u \GTSnegatedDirectedAdjacencyUp/ v$`

or `$u \GTSnegDirAdjUp/ v$`

or `$u \GTSnDAU/ v$`

will yield $u \nrightarrow v$.

`\GTSnegatedDirectedAdjacencyUpRelation`, `\GTSnegDirAdjUpRel`, `\GTSnDAUR` This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

`$u\GTSnegatedDirectedAdjacencyUpRelation/v$`

or `$u\GTSnegDirAdjUpRel/v$`

or `$u\GTSnDAUR/v$`

will yield $u \nrightarrow v$.

`\GTSdirectedAdjacencyDown`, `\GTSdirAdjDown`, `\GTSdAD` This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

`A.\GTSdirectedAdjacencyDown/A.`

or `A.\GTSdirAdjDown/A.`

or `A.\GTSnDAD/A.`

will yield $A \rightarrow A$.

`u \GTSdirectedAdjacencyDown/ v`

or `u \GTSdirAdjDown/ v`

or `u \GTSnDAD/ v`

will yield $u \rightarrow v$.

`$u \GTSdirectedAdjacencyDown/ v$`

or `$u \GTSdirAdjDown/ v$`

or `$u \GTSnDAD/ v$`

will yield $u \rightarrow v$.

`\GTSdirectedAdjacencyDownRelation`, `\GTSdirAdjDownRel`, `\GTSdADR` This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

`$u\GTSdirectedAdjacencyDownRelation/v$`

or `$u\GTSdirAdjDownRel/v$`

or `$u\GTSdADR/v$`

will yield $u \rightarrow v$.

`\GTSnegatedDirectedAdjacencyDown`, `\GTSnegDirAdjDown`, `\GTSnDAD` This macro will draw a negated directed adjacency downward symbol.

`A.\GTSnegatedDirectedAdjacencyDown/A.`

or `A.\GTSnegDirAdjDown/A.`

or `A.\GTSnDAD/A.`

will yield $A \nrightarrow A$.

`u \GTSnegatedDirectedAdjacencyDown/ v`

or $\boxed{u \ \backslash\text{GTSnegDirAdjDown}/ \ v}$

or $\boxed{u \ \backslash\text{GTSnDAD}/ \ v}$

will yield $\boxed{u \not\propto v}$.

$\boxed{\$u \ \backslash\text{GTSnegatedDirectedAdjacencyDown}/ \ v\$}$

or $\boxed{\$u \ \backslash\text{GTSnegDirAdjDown}/ \ v\$}$

or $\boxed{\$u \ \backslash\text{GTSnDAD}/ \ v\$}$

will yield $\boxed{u \not\propto v}$.

$\backslash\text{GTSnegatedDirectedAdjacencyDownRelation}$, $\backslash\text{GTSnegDirAdjDownRel}$, $\backslash\text{GTSnDADR}$ This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

$\boxed{\$u \backslash\text{GTSnegatedDirectedAdjacencyDownRelation}/v\$}$

or $\boxed{\$u \backslash\text{GTSnegDirAdjDownRel}/v\$}$

or $\boxed{\$u \backslash\text{GTSnDADR}/v\$}$

will yield $\boxed{u \not\propto v}$.

2.2.3 Standard incidence macros

$\backslash\text{GTS@i}$ This macro will draw any of the incidence symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and $\backslash\text{GTS@j}$: $\backslash\text{GTSxincidence}$, $\backslash\text{GTSxinc}$, and $\backslash\text{GTSxi}$. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

- $\{ \langle \text{mainColor} \rangle \}$ If non-empty, this color will be used as main color.
- $\{ \langle \text{secondColor} \rangle \}$ If non-empty, this color will be used as second color.
- $\{ \langle \text{drawSlash} \rangle \}$ Non-empty implies that a negation will be drawn.
- $\{ \langle \text{drawUpArrow} \rangle \}$ Non-empty implies that an up-arrow will be drawn.
- $\{ \langle \text{drawDownArrow} \rangle \}$ Non-empty implies that a down-arrow will be drawn.

$\backslash\text{GTS@j}$ This macro builds on top of $\backslash\text{GTS@i}$ to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and $\backslash\text{GTS@i}$: $\backslash\text{GTSxincidence}$, $\backslash\text{GTSxinc}$, and $\backslash\text{GTSxi}$.

Arguments:

- $\{ \langle \text{mainColor} \rangle \}$ If non-empty, this color will be used as main color.
- $\{ \langle \text{secondColor} \rangle \}$ If non-empty, this color will be used as second color.
- $\{ \langle \text{drawSlash} \rangle \}$ Non-empty implies that a negation will be drawn.
- $\{ \langle \text{drawUpArrow} \rangle \}$ Non-empty implies that an up-arrow will be drawn.
- $\{ \langle \text{drawDownArrow} \rangle \}$ Non-empty implies that a down-arrow will be drawn.
- $\{ \langle \text{relation} \rangle \}$ Non-empty implies that relation spacing will be added.

$\backslash\text{GTSxincidence}$, $\backslash\text{GTSxinc}$, $\backslash\text{GTSxi}$ Versatile macros for incidence that can be given options.

Argument:

$[\langle \text{optionsList} \rangle]$ Sets the following options if not empty:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
in or i,
out or o,
relation or rel or r.

$\backslash\text{GTSincidence}$, $\backslash\text{GTSinc}$, $\backslash\text{GTSi}$ This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the

symbol is incident.

$A.\backslash\text{GTSincidence}/A.$

or $A.\backslash\text{GTSinc}/A.$

or $A.\backslash\text{GTSi}/A.$

will yield $A.\downarrow A.$

$v \backslash\text{GTSincidence}/ e$

or $v \backslash\text{GTSinc}/ e$

or $v \backslash\text{GTSi}/ e$

will yield $v \downarrow e$

$\$v \backslash\text{GTSincidence}/ e\$$

or $\$v \backslash\text{GTSinc}/ e\$$

or $\$v \backslash\text{GTSi}/ e\$$

will yield $v \downarrow e$

$\backslash\text{GTSincidenceRelation}$, $\backslash\text{GTSincRel}$, $\backslash\text{GTSiR}$ This macro will draw an incidence symbol with relation spacing in math mode.

$\$v\backslash\text{GTSincidenceRelation}/e\$$

or $\$v\backslash\text{GTSincRel}/e\$$

or $\$v\backslash\text{GTSiR}/e\$$

will yield $v \downarrow e$

$\backslash\text{GTSnegatedIncidence}$, $\backslash\text{GTSnegInc}$, $\backslash\text{GTSnI}$ This macro will draw a negated incidence symbol.

$A.\backslash\text{GTSnegatedIncidence}/A.$

or $A.\backslash\text{GTSnegInc}/A.$

or $A.\backslash\text{GTSnI}/A.$

will yield $A.\not\downarrow A.$

$v \backslash\text{GTSnegatedIncidence}/ e$

or $v \backslash\text{GTSnegInc}/ e$

or $v \backslash\text{GTSnI}/ e$

will yield $v \not\downarrow e$

$\$v \backslash\text{GTSnegatedIncidence}/ e\$$

or $\$v \backslash\text{GTSnegInc}/ e\$$

or $\$v \backslash\text{GTSnI}/ e\$$

will yield $v \not\downarrow e$

$\backslash\text{GTSnegatedIncidenceRelation}$, $\backslash\text{GTSnegIncRel}$, $\backslash\text{GTSnIR}$ This macro will draw a negated incidence symbol with relation spacing in math mode.

$\$v\backslash\text{GTSnegatedIncidenceRelation}/e\$$

or $\$v\backslash\text{GTSnegIncRel}/e\$$

or $\$v\backslash\text{GTSnIR}/e\$$

will yield $v \not\downarrow e$

\GTSoutIncidence, \GTSoutInc, \GTSOI This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

`A.\GTSoutIncidence/A.`

or `A.\GTSoutInc/A.`

or `A.\GTSOI/A.`

will yield $A \overset{\circ}{\rightarrow} A.$

`v \GTSoutIncidence/ e`

or `v \GTSoutInc/ e`

or `v \GTSOI/ e`

will yield $v \overset{\circ}{\rightarrow} e.$

`$v \GTSoutIncidence/ e$`

or `$v \GTSoutInc/ e$`

or `$v \GTSOI/ e$`

will yield $v \overset{\circ}{\rightarrow} e.$

\GTSoutIncidenceRelation, \GTSoutIncRel, \GTSOIR This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

`$v\GTSoutIncidenceRelation/e$`

or `$v\GTSoutIncRel/e$`

or `$v\GTSOIR/e$`

will yield $v \overset{\circ}{\rightarrow} e.$

\GTSnegatedOutIncidence, \GTSnegOutInc, \GTSnOI This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

`A.\GTSnegatedOutIncidence/A.`

or `A.\GTSnegOutInc/A.`

or `A.\GTSnOI/A.`

will yield $A \overset{\circ}{\nrightarrow} A.$

`v \GTSnegatedOutIncidence/ e`

or `v \GTSnegOutInc/ e`

or `v \GTSnOI/ e`

will yield $v \overset{\circ}{\nrightarrow} e.$

`$v \GTSnegatedOutIncidence/ e$`

or `$v \GTSnegOutInc/ e$`

or `$v \GTSnOI/ e$`

will yield $v \overset{\circ}{\nrightarrow} e.$

\GTSnegatedOutIncidenceRelation, \GTSnegOutIncRel, \GTSnOIR This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

`$v\GTSnegatedOutIncidenceRelation/e$`

or `$v\GTSnegOutIncRel/e$`

or `$v\GTSnOIR/e$`

will yield $v \nrightarrow e$.

\GTSinIncidence, \GTSinInc, \GTSiI This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSinIncidence}/A.$

or $A.\backslash\text{GTSinInc}/A.$

or $A.\backslash\text{GTSiI}/A.$

will yield $A.\nrightarrow A.$

$v \backslash\text{GTSinIncidence}/ e$

or $v \backslash\text{GTSinInc}/ e$

or $v \backslash\text{GTSiI}/ e$

will yield $v \nrightarrow e$.

$\$v \backslash\text{GTSinIncidence}/ e\$$

or $\$v \backslash\text{GTSinInc}/ e\$$

or $\$v \backslash\text{GTSiI}/ e\$$

will yield $v \nrightarrow e$.

\GTSinIncidenceRelation, \GTSinIncRel, \GTSiIR This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSinIncidenceRelation}/e\$$

or $\$v\backslash\text{GTSinIncRel}/e\$$

or $\$v\backslash\text{GTSiIR}/e\$$

will yield $v \nrightarrow e$.

\GTSnegatedInIncidence, \GTSnegInInc, \GTSnII This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

$A.\backslash\text{GTSnegatedInIncidence}/A.$

or $A.\backslash\text{GTSnegInInc}/A.$

or $A.\backslash\text{GTSnII}/A.$

will yield $A.\nrightarrow A.$

$v \backslash\text{GTSnegatedInIncidence}/ e$

or $v \backslash\text{GTSnegInInc}/ e$

or $v \backslash\text{GTSnII}/ e$

will yield $v \nrightarrow e$.

$\$v \backslash\text{GTSnegatedInIncidence}/ e\$$

or $\$v \backslash\text{GTSnegInInc}/ e\$$

or $\$v \backslash\text{GTSnII}/ e\$$

will yield $v \nrightarrow e$.

\GTSnegatedInIncidenceRelation, \GTSnegInIncRel, \GTSnIIR This macro will draw a

negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`$v\GTSnegatedInIncidenceRelation/e$`

or `$v\GTSnegInIncRel/e$`

or `$v\GTSnIIR/e$`

will yield $v \nrightarrow e$.

2.2.4 Monochrome adjacency macros

`\GTSadjacencyMonochrome`, `\GTSadjMono`, `\GTSadjMon`, `\GTSaM` This macro will draw a monochrome adjacency symbol.

`A.\GTSadjacencyMonochrome/A.`

or `A.\GTSadjMono/A.`

or `A.\GTSadjMon/A.`

or `A.\GTSaM/A.`

will yield $A \circ A$.

`u \GTSadjacencyMonochrome/ v`

or `u \GTSadjMono/ v`

or `u \GTSadjMon/ v`

or `u \GTSaM/ v`

will yield $u \circ v$.

`$u \GTSadjacencyMonochrome/ v$`

or `$u \GTSadjMono/ v$`

or `$u \GTSadjMon/ v$`

or `$u \GTSaM/ v$`

will yield $u \circ v$.

`\GTSadjacencyMonochromeRelation`, `\GTSadjMonoRel`, `\GTSadjMonRel`, `\GTSaMR` This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

`$u\GTSadjacencyMonochromeRelation/v$`

or `$u\GTSadjMonoRel/v$`

or `$u\GTSadjMonRel/v$`

or `$u\GTSaMR/v$`

will yield $u \circ v$.

`\GTSunadjacencyMonochrome`, `\GTSunadjMono`, `\GTSunadjMon`, `\GTSuM` This macro will draw a monochrome unadjacency symbol.

`A.\GTSunadjacencyMonochrome/A.`

or `A.\GTSunadjMono/A.`

or `A.\GTSunadjMon/A.`

or `A.\GTSuM/A.`

will yield $A \circ A$.

`u \GTSunadjacencyMonochrome/ v`

or $u \text{ \textcircled{G} } v$

or $u \text{ \textcircled{G} } v$

or $u \text{ \textcircled{G} } v$

will yield $u \text{ \textcircled{G} } v$.

$\$u \text{ \textcircled{G} } v$

or $\$u \text{ \textcircled{G} } v$

or $\$u \text{ \textcircled{G} } v$

or $\$u \text{ \textcircled{G} } v$

will yield $u \text{ \textcircled{G} } v$.

\GTSunadjacencyMonochromeRelation, \GTSunadjMonoRel, \GTSunadjMonRel, \GTSuMR This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

$\$u \text{ \textcircled{G} } v$

or $\$u \text{ \textcircled{G} } v$

or $\$u \text{ \textcircled{G} } v$

or $\$u \text{ \textcircled{G} } v$

will yield $u \text{ \textcircled{G} } v$.

\GTSdirectedAdjacencyUpMonochrome, \GTSdirAdjUpMono, \GTSdirAdjUpMon, \GTSdAUM This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

$A \text{ \textcircled{G} } A$

or $A \text{ \textcircled{G} } A$

or $A \text{ \textcircled{G} } A$

or $A \text{ \textcircled{G} } A$

will yield $A \text{ \textcircled{G} } A$.

$u \text{ \textcircled{G} } v$

or $u \text{ \textcircled{G} } v$

or $u \text{ \textcircled{G} } v$

or $u \text{ \textcircled{G} } v$

will yield $u \text{ \textcircled{G} } v$.

$\$u \text{ \textcircled{G} } v$

or $\$u \text{ \textcircled{G} } v$

or $\$u \text{ \textcircled{G} } v$

or $\$u \text{ \textcircled{G} } v$

will yield $u \text{ \textcircled{G} } v$.

\GTSdirectedAdjacencyUpMonochromeRelation, \GTSdirAdjUpMonoRel, \GTSdirAdjUpMonRel, \GTSdAUMR This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

$\$u\backslash\text{GTSdirectedAdjacencyUpMonochromeRelation}/v\$$
 or $\$u\backslash\text{GTSdirAdjUpMonoRel}/v\$$
 or $\$u\backslash\text{GTSdirAdjUpMonRel}/v\$$
 or $\$u\backslash\text{GTSdAUMR}/v\$$
 will yield $u \overset{\circ}{\underset{\circ}{\rightharpoonup}} v$.

\GTSnegatedDirectedAdjacencyUpMonochrome, \GTSnegDirAdjUpMono, \GTSnegDirAdjUpMon, \GTSnDAUM This macro will draw a monochrome negated directed adjacency upward symbol.

$A.\backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/A.$
 or $A.\backslash\text{GTSnegDirAdjUpMono}/A.$
 or $A.\backslash\text{GTSnegDirAdjUpMon}/A.$
 or $A.\backslash\text{GTSnDAUM}/A.$
 will yield $A.\overset{\circ}{\underset{\circ}{\rightharpoonup}}A.$

$u \backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/ v$
 or $u \backslash\text{GTSnegDirAdjUpMono}/ v$
 or $u \backslash\text{GTSnegDirAdjUpMon}/ v$
 or $u \backslash\text{GTSnDAUM}/ v$
 will yield $u \overset{\circ}{\underset{\circ}{\rightharpoonup}} v$.

$\$u \backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/ v\$$
 or $\$u \backslash\text{GTSnegDirAdjUpMono}/ v\$$
 or $\$u \backslash\text{GTSnegDirAdjUpMon}/ v\$$
 or $\$u \backslash\text{GTSnDAUM}/ v\$$
 will yield $u \overset{\circ}{\underset{\circ}{\rightharpoonup}} v$.

\GTSnegatedDirectedAdjacencyUpMonochromeRelation, \GTSnegDirAdjUpMonoRel, \GTSnegDirAdjUpMonRel, \GTSnDAUMR This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

$\$u\backslash\text{GTSnegatedDirectedAdjacencyUpMonochromeRelation}/v\$$
 or $\$u\backslash\text{GTSnegDirAdjUpMonoRel}/v\$$
 or $\$u\backslash\text{GTSnegDirAdjUpMonRel}/v\$$
 or $\$u\backslash\text{GTSnDAUMR}/v\$$
 will yield $u \overset{\circ}{\underset{\circ}{\rightharpoonup}} v$.

\GTSdirectedAdjacencyDownMonochrome, \GTSdirAdjDownMono, \GTSdirAdjDownMon, \GTSdADM This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

$A.\backslash\text{GTSdirectedAdjacencyDownMonochrome}/A.$
 or $A.\backslash\text{GTSdirAdjDownMono}/A.$
 or $A.\backslash\text{GTSdirAdjDownMon}/A.$

or $A.\text{\textbackslash GTSdADM}/A.$

will yield $A.\text{\textcircled{g}}A.$

$u \text{\textbackslash GTSdirectedAdjacencyDownMonochrome}/ v$

or $u \text{\textbackslash GTSdirAdjDownMono}/ v$

or $u \text{\textbackslash GTSdirAdjDownMon}/ v$

or $u \text{\textbackslash GTSdADM}/ v$

will yield $u \text{\textcircled{g}} v.$

$\$u \text{\textbackslash GTSdirectedAdjacencyDownMonochrome}/ v\$$

or $\$u \text{\textbackslash GTSdirAdjDownMono}/ v\$$

or $\$u \text{\textbackslash GTSdirAdjDownMon}/ v\$$

or $\$u \text{\textbackslash GTSdADM}/ v\$$

will yield $u \text{\textcircled{g}} v.$

$\text{\textbackslash GTSdirectedAdjacencyDownMonochromeRelation}$, $\text{\textbackslash GTSdirAdjDownMonoRel}$, $\text{\textbackslash GTSdirAdjDownMonRel}$, $\text{\textbackslash GTSdADMR}$ This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

$\$u \text{\textbackslash GTSdirectedAdjacencyDownMonochromeRelation}/ v\$$

or $\$u \text{\textbackslash GTSdirAdjDownMonoRel}/ v\$$

or $\$u \text{\textbackslash GTSdirAdjDownMonRel}/ v\$$

or $\$u \text{\textbackslash GTSdADMR}/ v\$$

will yield $u \text{\textcircled{g}} v.$

$\text{\textbackslash GTSnegatedDirectedAdjacencyDownMonochrome}$, $\text{\textbackslash GTSnegDirAdjDownMono}$, $\text{\textbackslash GTSnegDirAdjDownMon}$, $\text{\textbackslash GTSnDADM}$ This macro will draw a monochrome negated directed adjacency downward symbol.

$A.\text{\textbackslash GTSnegatedDirectedAdjacencyDownMonochrome}/ A.$

or $A.\text{\textbackslash GTSnegDirAdjDownMono}/ A.$

or $A.\text{\textbackslash GTSnegDirAdjDownMon}/ A.$

or $A.\text{\textbackslash GTSnDADM}/ A.$

will yield $A.\text{\textcircled{g}}A.$

$u \text{\textbackslash GTSnegatedDirectedAdjacencyDownMonochrome}/ v$

or $u \text{\textbackslash GTSnegDirAdjDownMono}/ v$

or $u \text{\textbackslash GTSnegDirAdjDownMon}/ v$

or $u \text{\textbackslash GTSnDADM}/ v$

will yield $u \text{\textcircled{g}} v.$

$\$u \text{\textbackslash GTSnegatedDirectedAdjacencyDownMonochrome}/ v\$$

or $\$u \text{\textbackslash GTSnegDirAdjDownMono}/ v\$$

or $\$u \text{\textbackslash GTSnegDirAdjDownMon}/ v\$$

or $\$u \text{\textbackslash GTSnDADM}/ v\$$

will yield $u \text{\textcircled{g}} v.$

$\text{\textbackslash GTSnegatedDirectedAdjacencyDownMonochromeRelation}$, $\text{\textbackslash GTSnegDirAdjDownMonoRel}$,

\GTSnegDirAdjDownMonRel, \GTSnDADMR This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

$\$u\backslash\text{GTSnegatedDirectedAdjacencyDownMonochromeRelation}/v\$$
 or $\$u\backslash\text{GTSnegDirAdjDownMonoRel}/v\$$
 or $\$u\backslash\text{GTSnegDirAdjDownMonRel}/v\$$
 or $\$u\backslash\text{GTSnDADMR}/v\$$
 will yield $u \nrightarrow v$.

2.2.5 Monochrome incidence macros

\GTSincidenceMonochrome, \GTSincMono, \GTSincMon, \GTSiM This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSincidenceMonochrome}/A.$
 or $A.\backslash\text{GTSincMono}/A.$
 or $A.\backslash\text{GTSincMon}/A.$
 or $A.\backslash\text{GTSiM}/A.$
 will yield $A \circ A$.

$v \backslash\text{GTSincidenceMonochrome}/ e$
 or $v \backslash\text{GTSincMono}/ e$
 or $v \backslash\text{GTSincMon}/ e$
 or $v \backslash\text{GTSiM}/ e$
 will yield $v \circ e$.

$\$v \backslash\text{GTSincidenceMonochrome}/ e\$$
 or $\$v \backslash\text{GTSincMono}/ e\$$
 or $\$v \backslash\text{GTSincMon}/ e\$$
 or $\$v \backslash\text{GTSiM}/ e\$$
 will yield $v \circ e$.

\GTSincidenceMonochromeRelation, \GTSincMonoRel, \GTSincMonRel, \GTSiMR This macro will draw a monochrome incidence symbol with relation spacing in math mode.

$\$v\backslash\text{GTSincidenceMonochromeRelation}/e\$$
 or $\$v\backslash\text{GTSincMonoRel}/e\$$
 or $\$v\backslash\text{GTSincMonRel}/e\$$
 or $\$v\backslash\text{GTSiMR}/e\$$
 will yield $v \circ e$.

\GTSnegatedIncidenceMonochrome, \GTSnegIncMono, \GTSnegIncMon, \GTSnIM This macro will draw a monochrome negated incidence symbol.

$A.\backslash\text{GTSnegatedIncidenceMonochrome}/A.$
 or $A.\backslash\text{GTSnegIncMono}/A.$
 or $A.\backslash\text{GTSnegIncMon}/A.$
 or $A.\backslash\text{GTSnIM}/A.$
 will yield $A \nabla A$.

`v \GTSnegatedIncidenceMonochrome/ e`

or `v \GTSnegIncMono/ e`

or `v \GTSnegIncMon/ e`

or `v \GTSnIM/ e`

will yield $v \nrightarrow e$.

`$v \GTSnegatedIncidenceMonochrome/ e$`

or `$v \GTSnegIncMono/ e$`

or `$v \GTSnegIncMon/ e$`

or `$v \GTSnIM/ e$`

will yield $v \nrightarrow e$.

\GTSnegatedIncidenceMonochromeRelation, \GTSnegIncMonoRel, \GTSnegIncMonRel, \GTSnIMR This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

`$v\GTSnegatedIncidenceMonochromeRelation/e$`

or `$v\GTSnegIncMonoRel/e$`

or `$v\GTSnegIncMonRel/e$`

or `$v\GTSnIMR/e$`

will yield $v \nrightarrow e$.

\GTSoutIncidenceMonochrome, \GTSoutIncMono, \GTSoutIncMon, \GTSoIM This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

`A.\GTSoutIncidenceMonochrome/A.`

or `A.\GTSoutIncMono/A.`

or `A.\GTSoutIncMon/A.`

or `A.\GTSoIM/A.`

will yield $A \rightarrow A$.

`v \GTSoutIncidenceMonochrome/ e`

or `v \GTSoutIncMono/ e`

or `v \GTSoutIncMon/ e`

or `v \GTSoIM/ e`

will yield $v \rightarrow e$.

`$v \GTSoutIncidenceMonochrome/ e$`

or `$v \GTSoutIncMono/ e$`

or `$v \GTSoutIncMon/ e$`

or `$v \GTSoIM/ e$`

will yield $v \rightarrow e$.

\GTSoutIncidenceMonochromeRelation, \GTSoutIncMonoRel, \GTSoutIncMonRel, \GTSoIMR This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSoutIncidenceMonochromeRelation}/e\$$

or $\$v\backslash\text{GTSoutIncMonoRel}/e\$$

or $\$v\backslash\text{GTSoutIncMonRel}/e\$$

or $\$v\backslash\text{GTSoIMR}/e\$$

will yield $v \nearrow e$.

\GTSnegatedOutIncidenceMonochrome, \GTSnegOutIncMono, \GTSnegOutIncMon, \GTSnOIM This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

$A.\backslash\text{GTSnegatedOutIncidenceMonochrome}/A.$

or $A.\backslash\text{GTSnegOutIncMono}/A.$

or $A.\backslash\text{GTSnegOutIncMon}/A.$

or $A.\backslash\text{GTSnOIM}/A.$

will yield $A.\nrightarrow A.$

$v \backslash\text{GTSnegatedOutIncidenceMonochrome}/ e$

or $v \backslash\text{GTSnegOutIncMono}/ e$

or $v \backslash\text{GTSnegOutIncMon}/ e$

or $v \backslash\text{GTSnOIM}/ e$

will yield $v \nrightarrow e$.

$\$v \backslash\text{GTSnegatedOutIncidenceMonochrome}/ e\$$

or $\$v \backslash\text{GTSnegOutIncMono}/ e\$$

or $\$v \backslash\text{GTSnegOutIncMon}/ e\$$

or $\$v \backslash\text{GTSnOIM}/ e\$$

will yield $v \nrightarrow e$.

\GTSnegatedOutIncidenceMonochromeRelation, \GTSnegOutIncMonoRel, \GTSnegOutIncMonRel, \GTSnOIMR This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSnegatedOutIncidenceMonochromeRelation}/e\$$

or $\$v\backslash\text{GTSnegOutIncMonoRel}/e\$$

or $\$v\backslash\text{GTSnegOutIncMonRel}/e\$$

or $\$v\backslash\text{GTSnOIMR}/e\$$

will yield $v \nrightarrow e$.

\GTSinIncidenceMonochrome, \GTSinIncMono, \GTSinIncMon, \GTSiIM This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSinIncidenceMonochrome}/A.$

or $A.\backslash\text{GTSinIncMono}/A.$

or $A.\backslash\text{GTSinIncMon}/A.$

or $A.\text{\GTSiIM}/A.$

will yield $A.\text{\GTSiIM}A.$

$v \text{\GTSinIncidenceMonochrome}/ e$

or $v \text{\GTSinIncMono}/ e$

or $v \text{\GTSinIncMon}/ e$

or $v \text{\GTSiIM}/ e$

will yield $v \text{\GTSiIM} e.$

$\$v \text{\GTSinIncidenceMonochrome}/ e\$$

or $\$v \text{\GTSinIncMono}/ e\$$

or $\$v \text{\GTSinIncMon}/ e\$$

or $\$v \text{\GTSiIM}/ e\$$

will yield $v \text{\GTSiIM} e.$

\GTSinIncidenceMonochromeRelation, \GTSinIncMonoRel, \GTSinIncMonRel, \GTSiIMR This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

$\$v \text{\GTSinIncidenceMonochromeRelation}/ e\$$

or $\$v \text{\GTSinIncMonoRel}/ e\$$

or $\$v \text{\GTSinIncMonRel}/ e\$$

or $\$v \text{\GTSiIMR}/ e\$$

will yield $v \text{\GTSiIM} e.$

\GTSnegatedInIncidenceMonochrome, \GTSnegInIncMono, \GTSnegInIncMon, \GTSnIIM This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

$A.\text{\GTSnegatedInIncidenceMonochrome}/A.$

or $A.\text{\GTSnegInIncMono}/A.$

or $A.\text{\GTSnegInIncMon}/A.$

or $A.\text{\GTSnIIM}/A.$

will yield $A.\text{\GTSnIIM}A.$

$v \text{\GTSnegatedInIncidenceMonochrome}/ e$

or $v \text{\GTSnegInIncMono}/ e$

or $v \text{\GTSnegInIncMon}/ e$

or $v \text{\GTSnIIM}/ e$

will yield $v \text{\GTSnIIM} e.$

$\$v \text{\GTSnegatedInIncidenceMonochrome}/ e\$$

or $\$v \text{\GTSnegInIncMono}/ e\$$

or $\$v \text{\GTSnegInIncMon}/ e\$$

or $\$v \text{\GTSnIIM}/ e\$$

will yield $v \text{\GTSnIIM} e.$

\GTSnegatedInIncidenceMonochromeRelation, \GTSnegInIncMonoRel, \GTSnegInIncMonRel, \GTSnIIMR This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSnegatedInIncidenceMonochromeRelation}/e\$$
 or $\$v\backslash\text{GTSnegInIncMonoRel}/e\$$
 or $\$v\backslash\text{GTSnegInIncMonRel}/e\$$
 or $\$v\backslash\text{GTSnIIMR}/e\$$
 will yield $v \nrightarrow e$.

2.2.6 Meta-logic macros

\GTSMetaNot, \GTSMN This macro will draw a meta-not or relation-not symbol.

$A.\backslash\text{GTSMetaNot}/A.$
 or $A.\backslash\text{GTSMN}/A.$
 will yield $A.\mathrel{\mathop{\sqcap}}A.$
 $u \backslash\text{GTSMetaNot}/\backslash\text{GTSdAU}/ v$
 or $u \backslash\text{GTSMN}/\backslash\text{GTSdAU}/ v$
 will yield $u \mathrel{\mathop{\sqcap}} v.$
 $\$u \backslash\text{GTSMetaNot}/ \backslash\text{GTSdAU}/ v\$$
 or $\$u \backslash\text{GTSMN}/ \backslash\text{GTSdAU}/ v\$$
 will yield $u \mathrel{\mathop{\sqcap}} v.$

\GTSMetaAnd, \GTSMa This macro will draw a meta-and or relations-and symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

Look at the discussion after all logical symbols for less trivial uses and recommendations.

$A.\backslash\text{GTSMetaAnd}/A.$
 or $A.\backslash\text{GTSMa}/A.$
 will yield $A.\mathrel{\mathop{\sqcap}}A.$
 $u \backslash\text{GTSdAD}/\backslash\text{GTSMetaAnd}/\backslash\text{GTSdAU}/ v$
 or $u \backslash\text{GTSdAD}/\backslash\text{GTSMa}/\backslash\text{GTSdAU}/ v$
 will yield $u \mathrel{\mathop{\sqcap}} v.$
 $\$u \backslash\text{GTSdAD}/ \backslash\text{GTSMetaAnd}/ \backslash\text{GTSdAU}/ v\$$
 or $\$u \backslash\text{GTSdAD}/ \backslash\text{GTSMa}/ \backslash\text{GTSdAU}/ v\$$
 will yield $u \mathrel{\mathop{\sqcap}} v.$

\GTSMetaAndRelation, \GTSMetaAndRel, \GTSMaR This macro will draw a meta-and symbol with relation spacing in math mode.

$\$u \backslash\text{GTSdAD}/\backslash\text{GTSMetaAndRelation}/\backslash\text{GTSdAU}/ v\$$
 or $\$u \backslash\text{GTSdAD}/\backslash\text{GTSMetaAndRel}/\backslash\text{GTSdAU}/ v\$$
 or $\$u \backslash\text{GTSdAD}/\backslash\text{GTSMaR}/\backslash\text{GTSdAU}/ v\$$
 will yield $u \mathrel{\mathop{\sqcap}} v.$

\GTSMetaOr, \GTSMo This macro will draw a meta-or or relations-or symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

Look at the discussion after all logical symbols for less trivial uses and recommendations.

$A.\backslash\text{GTSMetaOr}/A.$
 or $A.\backslash\text{GTSMo}/A.$
 will yield $A.\mathrel{\mathop{\sqcup}}A.$

`u \GTSdAD/\GTSmetaOr/\GTSdAU/ v`
or `u \GTSdAD/\GTSmO/\GTSdAU/ v`
will yield $u \overset{\circ}{\underset{\circ}{\vee}} v$.
`$u \GTSdAD/ \GTSmetaOr/ \GTSdAU/ v$`
or `$u \GTSdAD/ \GTSmO/ \GTSdAU/ v$`
will yield $u \overset{\circ}{\underset{\circ}{\vee}} v$.

`\GTSmetaOrRelation, \GTSmetaOrRel, \GTSmOR` This macro will draw a meta-or symbol with relation spacing in math mode.
`$u \GTSdAD/\GTSmetaOrRelation/\GTSdAU/ v$`
or `$u \GTSdAD/\GTSmetaOrRel/\GTSdAU/ v$`
or `$u \GTSdAD/\GTSmOR/\GTSdAU/ v$`
will yield $u \overset{\circ}{\underset{\circ}{\vee}} v$.

`\GTSxmetallogic, \GTSxml` Versatile macros for meta-logic symbols that can be given options.

Argument:

`[(optionsList)]` Sets the following options if not empty:

not or n,
and or a,
or or o,
relation or rel or r.

Of course first three options are incompatible, and the fourth is compatible only with the second and third options.

`\GTSx` One macro to rule them all. Versatile macro for all symbols that can be given options, and a symbol type mandatory switch.

Argument:

`[(optionsList)]` Sets the following options if not empty:

- If `{symbolType}` below is a, see `\GTSxa`:
`mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,
`secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,
monochrome or mono or mon or m (equivalent to `mC=black,sC=black`),
negated or not or n,
up or u,
down or d,
relation or rel or r.
- If `{symbolType}` below is i, see `\GTSxi`:
`mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,
`secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,
monochrome or mono or mon or m (equivalent to `mC=black,sC=black`),
negated or not or n,
in or i,
out or o,
relation or rel or r.
- If `{symbolType}` below is m, see `\GTSxm1`:
not or n,
and or a,
or or o,
relation or rel or r.

`{symbolType}` a mandatory switch; it can take value a for adjacency, i for incidence, m for meta-logic.

2.3 Good practices

As is customary in algebra and logic, the conjunction/multiplication can be replaced by concatenation, and has higher priority than disjunction. So the following formulas are equivalent; we recommend using the shortest one:

Tournament adjacency:

$$\forall u, v, u \text{ --- } \overset{\text{m}}{\nabla} \text{ --- } v$$

is preferred over

$$\forall u, v, u \text{ } \begin{array}{c} \textcircled{\text{m}} \\ \text{---} \\ \textcircled{\wedge} \end{array} \begin{array}{c} \textcircled{\text{m}} \\ \text{---} \\ \textcircled{\vee} \end{array} \begin{array}{c} \textcircled{\text{m}} \\ \text{---} \\ \textcircled{\wedge} \end{array} \begin{array}{c} \textcircled{\text{m}} \\ \text{---} \\ \textcircled{\text{v}} \end{array} v,$$
$$\text{or } \forall u, v, u \left(\bigotimes_{\otimes}^m \bigwedge \bigotimes \right) \bigvee \left(\bigotimes_{\otimes}^m \bigwedge \bigotimes \right) v,$$

or even $\forall u, v, ((u \overset{\circ}{\times} v) \wedge (u \overset{\circ}{\times} v)) \vee ((u \overset{\circ}{\times} v) \wedge (u \overset{\circ}{\times} v))$.

Note the use of macros without Relation suffix above, and adding a single `\mathrel{}` around the composition of all relations:

$$\backslash\mathrm{mathrel}\{\backslash\mathrm{GTSdAD}/\backslash\mathrm{GTSnDAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSdAU}/\}.$$

This is the spacing practice that we recommend, although for very long compositions, it may be quite unreadable, and then spacing everything with `Relation` could/should be preferred.

We also recommend using brackets when the composition reflects all of the adjacencies between two vertices:

Tournament adjacency:

$$\forall u, v, u \left[\begin{array}{c} \textcircled{\circ} \textcircled{\circ} \textcircled{\circ} \\ \textcircled{\circ} \textcircled{\circ} \textcircled{\circ} \end{array} \textcircled{\circ} \textcircled{\circ} \right] v.$$

Tournament adjacency with maybe some colored edge also, who knows?:

$$\forall u, v, u \text{ } \begin{array}{c} \textcircled{\circ} \textcircled{\circ} \textcircled{\circ} \\ \textcircled{\circ} \textcircled{\circ} \textcircled{\circ} \end{array} \text{ } \textcircled{\circ} \textcircled{\circ} \textcircled{\circ} \text{ } v.$$

Directed graph:

[illegible]

Oriented graph:

[illegible]

Finally, we recommend prefix notation without “big” symbol for a sequence of consecutive ors or ands (with parentheses if needed):

$$\exists u, v, u \left[\overset{\text{m}}{\underset{\text{V}}{\vee}} \begin{array}{ccc} \text{red} & \text{green} & \text{blue} \\ \text{red} & \text{green} & \text{blue} \end{array} \right] v.$$
$$\exists v, e, v \left[\overset{\text{m}}{\underset{\text{V}}{\text{V}}} \text{ } \text{ } \text{ } \right] e.$$

3 Implementation

```

1 (*package)
2 \RequirePackage{amsmath}
3 \RequirePackage{amssymb}
4 \RequirePackage{graphicx}
5 \RequirePackage{keyval}
6 \RequirePackage{fdsymbol}
7 \RequirePackage{lua-unicode-math}
8 \RequirePackage{xcolor}
9
10 \def\GTS@mainColor{black}
11 \def\GTS@secondColor{gray}
12 \def\GTS@mainColorBackup{black}
13 \def\GTS@secondColorBackup{gray}
14 \def\GTS@drawSlash{}
15 \def\GTS@drawUpArrow{}
16 \def\GTS@drawDownArrow{}
17

```

\GTS@newcommand This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`\{<commandname>\}` The name of the command to create.

$\{\langle commandcode \rangle\}$ The code of the command to create.

```
18 \@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
19   \@ifdefinable{#1}{\def#1/#2}}
```

20 }}

\GTS@newprotectedcommand This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

{<commandname>} The name of the command to create.

{<commandcode>} The code of the command to create.

```
21 \ifdefinable{\GTS@newprotectedcommand}{\def\GTS@newprotectedcommand#1#2{%
22   \ifdefinable{#1}{\protected\def#1/{#2}}
23 }}
```

\GTSsetMainColor This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

{<color>} The color that must be used by GraphTheorySymbol afterward.

```
24 \newcommand{\GTSsetMainColor}[1]{%
25 \edef\GTS@mainColorTemp{#1}%
26 \edef\GTS@mainColorBackup{\GTS@mainColor}%
27 \edef\GTS@mainColor{\GTS@mainColorTemp}%
28 }
```

\GTSsetSecondColor This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

{<color>} The color that must be used by GraphTheorySymbol afterward.

```
29 \newcommand{\GTSsetSecondColor}[1]{%
30 \edef\GTS@secondColorTemp{#1}%
31 \edef\GTS@secondColorBackup{\GTS@secondColor}%
32 \edef\GTS@secondColor{\GTS@secondColorTemp}%
33 }
```

\GTS@a This macro will draw any adjacency symbol given the correct parameters. Its use is purely technical. See the macros derived from it and \GTS@b: \GTSxadjacency, \GTSxadj, and \GTSxa.

Arguments:

{<mainColor>} If non-empty, this color will be used as main color.

{<secondColor>} If non-empty, this color will be used as second color.

{<drawSlash>} Non-empty implies that a negation will be drawn.

{<drawUpArrow>} Non-empty implies that an up-arrow will be drawn.

{<drawDownArrow>} Non-empty implies that a down-arrow will be drawn.

```
34 \newcommand{\GTS@a}[5]{%
35 \if#1\empty\else%
36 \GTSsetMainColor{#1}%
37 \fi%
38 \if#2\empty\else%
39 \GTSsetSecondColor{#2}%
40 \fi%
41 \text{\ensuremath{%
42 \raisebox{-0.32ex}{\scalebox{1.5}[1.5]{%
43 %
44 \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
45 $\mathcolor{\GTS@secondColor}{\circ}$%
46 }}%
47 \hspace{-0.39ex}%
48 \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
49 $\mathcolor{\GTS@secondColor}{\bullet}$%
50 }}%
51 \hspace{-0.12ex}%
52 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
53 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
54 }}%
55 }}
```

```

55 \hspace{-0.40ex}%
56 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
57 $\mathcolor{\GTS@secondColor}{\circ}$%
58 }}%
59 \hspace{-0.39ex}%
60 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
61 $\mathcolor{\GTS@secondColor}{\bullet}$%
62 }}%
63 \hspace{0.27ex}%
if GTS@drawUpArrow empty else
64 \if#4\empty\else%
65 \hspace{-0.65ex}%
66 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
67 \textcolor{\GTS@mainColor}{\textasciicircum}%
68 }}%
69 \fi%
if GTS@drawDownArrow empty else
70 \if#5\empty\else%
71 \hspace{-0.65ex}%
72 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
73 \textcolor{\GTS@mainColor}{\textasciicircum}%
74 }}%
75 \fi%
if GTS@drawSlash empty else
76 \if#3\empty\else%
77 \hspace{-0.65ex}%
78 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
79 \textcolor{black}{/}%
80 }}%
81 \fi%
82 %
83 }}%
84 }}%
85 \if#1\empty\else%
86 \GTSsetMainColor{\GTS@mainColorBackup}%
87 \fi%
88 \if#2\empty\else%
89 \GTSsetSecondColor{\GTS@secondColorBackup}%
90 \fi%
91 }

```

\GTS@b This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and **\GTS@a**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

{\mainColor}} If non-empty, this color will be used as main color.
{\secondColor}} If non-empty, this color will be used as second color.
{\drawSlash}} Non-empty implies that a negation will be drawn.
{\drawUpArrow}} Non-empty implies that an up-arrow will be drawn.
{\drawDownArrow}} Non-empty implies that a down-arrow will be drawn.
{\relation}} Non-empty implies that relation spacing will be added.

```

92 \newcommand{\GTS@b}[6]{%
93 \if#61%
94 \mathrel{\GTS@a{#1}{#2}{#3}{#4}{#5}}%
95 \else%
96 \GTS@a{#1}{#2}{#3}{#4}{#5}%
97 \fi%
98 }

```

\GTSxadjacency, **\GTSxadj**, **\GTSxa** Versatile macros for adjacency that can be given options.

Argument:

[*(optionsList)*] Sets the following options if not empty:

mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
up or u,
down or d,
relation or rel or r.

```

99 \define@key{GTSxa}{mainColor}{\def\GTS@@mainColor{#1}}
100 \define@key{GTSxa}{mainC}{\def\GTS@@mainColor{#1}}
101 \define@key{GTSxa}{mColor}{\def\GTS@@mainColor{#1}}
102 \define@key{GTSxa}{mC}{\def\GTS@@mainColor{#1}}
103 \define@key{GTSxa}{secondColor}{\def\GTS@@secondColor{#1}}
104 \define@key{GTSxa}{secondC}{\def\GTS@@secondColor{#1}}
105 \define@key{GTSxa}{sColor}{\def\GTS@@secondColor{#1}}
106 \define@key{GTSxa}{sC}{\def\GTS@@secondColor{#1}}
107 \define@key{GTSxa}{monochrome}[1]%
108 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
109 \define@key{GTSxa}{mono}[1]%
110 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
111 \define@key{GTSxa}{mon}[1]%
112 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
113 \define@key{GTSxa}{m}[1]%
114 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
115 \define@key{GTSxa}{negated}[1]{\def\GTS@@drawSlash{1}}
116 \define@key{GTSxa}{not}[1]{\def\GTS@@drawSlash{1}}
117 \define@key{GTSxa}{n}[1]{\def\GTS@@drawSlash{1}}
118 \define@key{GTSxa}{up}[1]{\def\GTS@@drawUpArrow{1}}
119 \define@key{GTSxa}{u}[1]{\def\GTS@@drawUpArrow{1}}
120 \define@key{GTSxa}{down}[1]{\def\GTS@@drawDownArrow{1}}
121 \define@key{GTSxa}{d}[1]{\def\GTS@@drawDownArrow{1}}
122 \define@key{GTSxa}{relation}[1]{\def\GTS@@relation{1}}
123 \define@key{GTSxa}{rel}[1]{\def\GTS@@relation{1}}
124 \define@key{GTSxa}{r}[1]{\def\GTS@@relation{1}}
125
126 \newcommand{\GTSxa}[1][{}]{%
127 \def\GTS@@mainColor{}%
128 \def\GTS@@secondColor{}%
129 \def\GTS@@drawSlash{}%
130 \def\GTS@@drawUpArrow{}%
131 \def\GTS@@drawDownArrow{}%
132 \def\GTS@@relation{}%
133 \setkeys{GTSxa}{#1}%
134 \GTS@b{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
135 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
136 }
137
138 \newcommand{\GTSxadj}[1][{}]{\GTSxa[#1]}
139
140 \newcommand{\GTSxadjacency}[1][{}]{\GTSxa[#1]}

```

\GTSadjacency, \GTSadj, \GTSa This macro will draw an adjacency symbol.

```

141 \GTS@newcommand{\GTSadjacency}{\GTS@a{}{}{}{}}
142 \GTS@newcommand{\GTSadj}{\GTS@a{}{}{}{}}
143 \GTS@newcommand{\GTSa}{\GTS@a{}{}{}{}}

```

\GTSadjacencyRelation, \GTSadjRel, \GTSaR This macro will draw an adjacency symbol with relation spacing in math mode.

```

144 \GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSa/}}
145 \GTS@newcommand{\GTSadjRel}{\mathrel{\GTSa/}}
146 \GTS@newcommand{\GTSaR}{\mathrel{\GTSa/}}

```

\GTSunadjacency, \GTSunadj, \GTSu This macro will draw an unadjacency symbol.

```
147 \GTS@newcommand{\GTSunadjacency}{\GTS@a{}{}{}{1}{}{}}
148 \GTS@newcommand{\GTSunadj}{\GTS@a{}{}{}{1}{}{}}
149 \GTS@newcommand{\GTSu}{\GTS@a{}{}{}{1}{}{}}
```

\GTSunadjacencyRelation, \GTSunadjRel, \GTSuR This macro will draw an unadjacency symbol with relation spacing in math mode.

```
150 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSu/}}
151 \GTS@newcommand{\GTSunadjRel}{\mathrel{\GTSu/}}
152 \GTS@newcommand{\GTSuR}{\mathrel{\GTSu/}}
```

\GTSdirectedAdjacencyUp, \GTSdirAdjUp, \GTSdAU This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```
153 \GTS@newcommand{\GTSdirectedAdjacencyUp}{\GTS@a{}{}{}{1}{}{}}
154 \GTS@newcommand{\GTSdirAdjUp}{\GTS@a{}{}{}{1}{}{}}
155 \GTS@newcommand{\GTSdAU}{\GTS@a{}{}{}{1}{}{}}
```

\GTSdirectedAdjacencyUpRelation, \GTSdirAdjUpRel, \GTSdAUR This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

```
156 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{\mathrel{\GTSdAU/}}
157 \GTS@newcommand{\GTSdirAdjUpRel}{\mathrel{\GTSdAU/}}
158 \GTS@newcommand{\GTSdAUR}{\mathrel{\GTSdAU/}}
```

\GTSnegatedDirectedAdjacencyUp, \GTSnegDirAdjUp, \GTSnDAU This macro will draw a negated directed adjacency upward symbol.

```
159 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{\GTS@a{}{}{}{1}{1}{}{}}
160 \GTS@newcommand{\GTSnegDirAdjUp}{\GTS@a{}{}{}{1}{1}{}{}}
161 \GTS@newcommand{\GTSnDAU}{\GTS@a{}{}{}{1}{1}{}{}}
```

\GTSnegatedDirectedAdjacencyUpRelation, \GTSnegDirAdjUpRel, \GTSnDAUR This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

```
162 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
163 \mathrel{\GTSnDAU/}%
164 }
165 \GTS@newcommand{\GTSnegDirAdjUpRel}{\mathrel{\GTSnDAU/}}
166 \GTS@newcommand{\GTSnDAUR}{\mathrel{\GTSnDAU/}}
```

\GTSdirectedAdjacencyDown, \GTSdirAdjDown, \GTSdAD This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```
167 \GTS@newcommand{\GTSdirectedAdjacencyDown}{\GTS@a{}{}{}{}{1}}
168 \GTS@newcommand{\GTSdirAdjDown}{\GTS@a{}{}{}{}{1}}
169 \GTS@newcommand{\GTSdAD}{\GTS@a{}{}{}{}{1}}
```

\GTSdirectedAdjacencyDownRelation, \GTSdirAdjDownRel, \GTSdADR This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

```
170 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{\mathrel{\GTSdAD/}}
171 \GTS@newcommand{\GTSdirAdjDownRel}{\mathrel{\GTSdAD/}}
172 \GTS@newcommand{\GTSdADR}{\mathrel{\GTSdAD/}}
```

\GTSnegatedDirectedAdjacencyDown, \GTSnegDirAdjDown, \GTSnDAD This macro will draw a negated directed adjacency downward symbol.

```
173 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{\GTS@a{}{}{}{}{1}}
174 \GTS@newcommand{\GTSnegDirAdjDown}{\GTS@a{}{}{}{}{1}}
175 \GTS@newcommand{\GTSnDAD}{\GTS@a{}{}{}{}{1}}
```

`\GTSnegatedDirectedAdjacencyDownRelation`, `\GTSnegDirAdjDownRel`, `\GTSnDADR` This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

```
176 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
177 \mathrel{\GTSnDAD/}%
178 }
179 \GTS@newcommand{\GTSnegDirAdjDownRel}{\mathrel{\GTSnDAD/}}
180 \GTS@newcommand{\GTSnDADR}{\mathrel{\GTSnDAD/}}
```

`\GTS@i` This macro will draw any incidence symbol given the correct parameters. Its use is purely technical. See the macros derived from it and `\GTS@j`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

`{\mainColor}` If non-empty, this color will be used as main color.
`{\secondColor}` If non-empty, this color will be used as second color.
`{\drawSlash}` Non-empty implies that a negation will be drawn.
`{\drawUpArrow}` Non-empty implies that an up-arrow will be drawn.
`{\drawDownArrow}` Non-empty implies that a down-arrow will be drawn.

```
181 \newcommand{\GTS@i}[5]{%
182 \if#1\empty\else%
183 \GTSsetMainColor{#1}%
184 \fi%
185 \if#2\empty\else%
186 \GTSsetSecondColor{#2}%
187 \fi%
188 \text{\ensuremath{%
189 \raisebox{-0.22ex}{\scalebox{2.05}[2.05]{%
190 %
191 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
192 $\mathcolor{\GTS@secondColor}{\circ}$%
193 }}}}
194 \hspace{-0.39ex}%
195 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
196 $\mathcolor{\GTS@secondColor}{\bullet}$%
197 }}}}
198 \hspace{-0.12ex}%
199 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
200 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
201 }}}}
202 \hspace{-0.18ex}%
203 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
204 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
205 }}}}
206 \hspace{0.2ex}%
if GTS@drawUpArrow empty else
207 \if#4\empty\else%
208 \hspace{-0.65ex}%
209 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
210 \textcolor{\GTS@secondColor}{\textasciicircum}%
211 }}}}
212 \fi%
if GTS@drawDownArrow empty else
213 \if#5\empty\else%
214 \hspace{-0.65ex}%
215 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
216 \textcolor{\GTS@secondColor}{\textasciicircum}%
217 }}}}
218 \fi%
```

```

if GTS@drawSlash empty else
219 \if#3\empty\else%
220 \hspace{-0.65ex}%
221 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
222 \textcolor{black}{/}%
223 }}%
224 \fi%
225 %
226 }}%
227 }}%
228 \if#1\empty\else%
229 \GTSsetMainColor{\GTS@mainColorBackup}%
230 \fi%
231 \if#2\empty\else%
232 \GTSsetSecondColor{\GTS@secondColorBackup}%
233 \fi%
234 }

```

\GTS@j This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and **\GTS@i**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**.

Arguments:

{\mainColor} If non-empty, this color will be used as main color.
{\secondColor} If non-empty, this color will be used as second color.
{\drawSlash} Non-empty implies that a negation will be drawn.
{\drawUpArrow} Non-empty implies that an up-arrow will be drawn.
{\drawDownArrow} Non-empty implies that a down-arrow will be drawn.
{\relation} Non-empty implies that relation spacing will be added.

```

235 \newcommand{\GTS@j}[6]{%
236 \if#61%
237 \mathrel{\GTS@i{#1}{#2}{#3}{#4}{#5}}%
238 \else%
239 \GTS@i{#1}{#2}{#3}{#4}{#5}%
240 \fi%
241 }

```

\GTSxincidence, **\GTSxinc**, **\GTSxi** Versatile macros for incidence that can be given options.

Argument:

[\optionsList] Sets the following options if not empty:

mainColor=color1 or **mainC=color1** or **mColor=color1** or **mC=color1**,
secondColor=color2 or **secondC=color2** or **sColor=color2** or **sC=color2**,
monochrome or **mono** or **mon** or **m** (equivalent to **mC=black,sC=black**),
negated or **not** or **n**,
in or **i**,
out or **o**,
relation or **rel** or **r**.

```

242 \define@key{\GTSxi}{mainColor}{\def\GTS@mainColor{#1}}
243 \define@key{\GTSxi}{mainC}{\def\GTS@mainColor{#1}}
244 \define@key{\GTSxi}{mColor}{\def\GTS@mainColor{#1}}
245 \define@key{\GTSxi}{mC}{\def\GTS@mainColor{#1}}
246 \define@key{\GTSxi}{secondColor}{\def\GTS@secondColor{#1}}
247 \define@key{\GTSxi}{secondC}{\def\GTS@secondColor{#1}}
248 \define@key{\GTSxi}{sColor}{\def\GTS@secondColor{#1}}
249 \define@key{\GTSxi}{sC}{\def\GTS@secondColor{#1}}
250 \define@key{\GTSxi}{monochrome}[1]%
251 {\def\GTS@mainColor{black}\def\GTS@secondColor{black}}
252 \define@key{\GTSxi}{mono}[1]%
253 {\def\GTS@mainColor{black}\def\GTS@secondColor{black}}
254 \define@key{\GTSxi}{mon}[1]%
255 {\def\GTS@mainColor{black}\def\GTS@secondColor{black}}
256 \define@key{\GTSxi}{m}[1]%

```

```

257 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
258 \define@key{GTSxi}{negated}[1]{\def\GTS@@drawSlash{1}}
259 \define@key{GTSxi}{not}[1]{\def\GTS@@drawSlash{1}}
260 \define@key{GTSxi}{n}[1]{\def\GTS@@drawSlash{1}}
261 \define@key{GTSxi}{out}[1]{\def\GTS@@drawUpArrow{1}}
262 \define@key{GTSxi}{o}[1]{\def\GTS@@drawUpArrow{1}}
263 \define@key{GTSxi}{in}[1]{\def\GTS@@drawDownArrow{1}}
264 \define@key{GTSxi}{i}[1]{\def\GTS@@drawDownArrow{1}}
265 \define@key{GTSxi}{relation}[1]{\def\GTS@@relation{1}}
266 \define@key{GTSxi}{rel}[1]{\def\GTS@@relation{1}}
267 \define@key{GTSxi}{r}[1]{\def\GTS@@relation{1}}
268
269 \newcommand{\GTSxi}[1][]{\%
270 \def\GTS@@mainColor{}%
271 \def\GTS@@secondColor{}%
272 \def\GTS@@drawSlash{}%
273 \def\GTS@@drawUpArrow{}%
274 \def\GTS@@drawDownArrow{}%
275 \def\GTS@@relation{}%
276 \setkeys{GTSxi}{#1}%
277 \GTS@j{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
278 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
279 }
280
281 \newcommand{\GTSxinc}[1][]{\GTSxi[#1]}
282
283 \newcommand{\GTSxincidence}[1][]{\GTSxi[#1]}

```

\GTSincidence, \GTSinc, \GTSi This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

284 \GTS@newcommand{\GTSincidence}{\GTS@i{}{}{}{}}
285 \GTS@newcommand{\GTSinc}{\GTS@i{}{}{}{}}
286 \GTS@newcommand{\GTSi}{\GTS@i{}{}{}{}}

```

\GTSincidenceRelation, \GTSincRel, \GTSiR This macro will draw an incidence symbol with relation spacing in math mode.

```

287 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSi/}}
288 \GTS@newcommand{\GTSincRel}{\mathrel{\GTSi/}}
289 \GTS@newcommand{\GTSiR}{\mathrel{\GTSi/}}

```

\GTSnegatedIncidence, \GTSnegInc, \GTSnI This macro will draw a negated incidence symbol.

```

290 \GTS@newcommand{\GTSnegatedIncidence}{\GTS@i{}{}{1}{}{}}
291 \GTS@newcommand{\GTSnegInc}{\GTS@i{}{}{1}{}{}}
292 \GTS@newcommand{\GTSnI}{\GTS@i{}{}{1}{}{}}

```

\GTSnegatedIncidenceRelation, \GTSnegIncRel, \GTSnIR This macro will draw a negated incidence symbol with relation spacing in math mode.

```

293 \GTS@newcommand{\GTSnegatedIncidenceRelation}{\mathrel{\GTSnI/}}
294 \GTS@newcommand{\GTSnegIncRel}{\mathrel{\GTSnI/}}
295 \GTS@newcommand{\GTSnIR}{\mathrel{\GTSnI/}}

```

\GTSoutIncidence, \GTSoutInc, \GTSoI This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

296 \GTS@newcommand{\GTSoutIncidence}{\GTS@i{}{}{1}{}{}}
297 \GTS@newcommand{\GTSoutInc}{\GTS@i{}{}{1}{}{}}
298 \GTS@newcommand{\GTSoI}{\GTS@i{}{}{1}{}{}}

```

\GTSoutIncidenceRelation, \GTSoutIncRel, \GTSOI This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
299 \GTS@newcommand{\GTSoutIncidenceRelation}{\mathrel{\GTSOI/}}
300 \GTS@newcommand{\GTSoutIncRel}{\mathrel{\GTSOI/}}
301 \GTS@newcommand{\GTSOI}{\mathrel{\GTSOI/}}
```

\GTSnegatedOutIncidence, \GTSnegOutInc, \GTSnOI This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

```
302 \GTS@newcommand{\GTSnegatedOutIncidence}{\GTS@i{}{}{1}{1}{}}
303 \GTS@newcommand{\GTSnegOutInc}{\GTS@i{}{}{1}{1}{}}
304 \GTS@newcommand{\GTSnOI}{\GTS@i{}{}{1}{1}{}}
```

\GTSnegatedOutIncidenceRelation, \GTSnegOutIncRel, \GTSnOIR This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
305 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{\mathrel{\GTSnOI/}}
306 \GTS@newcommand{\GTSnegOutIncRel}{\mathrel{\GTSnOI/}}
307 \GTS@newcommand{\GTSnOIR}{\mathrel{\GTSnOI/}}
```

\GTSinIncidence, \GTSinInc, \GTSiI This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
308 \GTS@newcommand{\GTSinIncidence}{\GTS@i{}{}{}{1}}
309 \GTS@newcommand{\GTSinInc}{\GTS@i{}{}{}{1}}
310 \GTS@newcommand{\GTSiI}{\GTS@i{}{}{}{1}}
```

\GTSinIncidenceRelation, \GTSinIncRel, \GTSiIR This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
311 \GTS@newcommand{\GTSinIncidenceRelation}{\mathrel{\GTSiI/}}
312 \GTS@newcommand{\GTSinIncRel}{\mathrel{\GTSiI/}}
313 \GTS@newcommand{\GTSiIR}{\mathrel{\GTSiI/}}
```

\GTSnegatedInIncidence, \GTSnegInInc, \GTSnII This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

```
314 \GTS@newcommand{\GTSnegatedInIncidence}{\GTS@i{}{}{1}{1}{}}
315 \GTS@newcommand{\GTSnegInInc}{\GTS@i{}{}{1}{1}{}}
316 \GTS@newcommand{\GTSnII}{\GTS@i{}{}{1}{1}{}}
```

\GTSnegatedInIncidenceRelation, \GTSnegInIncRel, \GTSnIIR This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
317 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{\mathrel{\GTSnII/}}
318 \GTS@newcommand{\GTSnegInIncRel}{\mathrel{\GTSnII/}}
319 \GTS@newcommand{\GTSnIIR}{\mathrel{\GTSnII/}}
```

\GTSadjacencyMonochrome, \GTSadjMono, \GTSadjMon, \GTSaM This macro will draw a monochrome adjacency symbol.

```
320 \GTS@newcommand{\GTSadjacencyMonochrome}{\GTS@a{black}{black}{}}{}{}
321 \GTS@newcommand{\GTSadjMono}{\GTS@a{black}{black}{}}{}{}
322 \GTS@newcommand{\GTSadjMon}{\GTS@a{black}{black}{}}{}{}
323 \GTS@newcommand{\GTSaM}{\GTS@a{black}{black}{}}{}{}

```

\GTSadjacencyMonochromeRelation, \GTSadjMonoRel, \GTSadjMonRel, \GTSaMR This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

```
324 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{\mathrel{\GTSaM/}}
325 \GTS@newcommand{\GTSadjMonoRel}{\mathrel{\GTSaM/}}
326 \GTS@newcommand{\GTSadjMonRel}{\mathrel{\GTSaM/}}
327 \GTS@newcommand{\GTSaMR}{\mathrel{\GTSaM/}}
```

\GTSunadjacencyMonochrome, \GTSunadjMono, \GTSunadjMon, \GTSuM This macro will draw a monochrome unadjacency symbol.

```
328 \GTS@newcommand{\GTSunadjacencyMonochrome}{\GTS@a{black}{black}{1}{}}
329 \GTS@newcommand{\GTSunadjMono}{\GTS@a{black}{black}{1}{}}
330 \GTS@newcommand{\GTSunadjMon}{\GTS@a{black}{black}{1}{}}
331 \GTS@newcommand{\GTSuM}{\GTS@a{black}{black}{1}{}}
```

\GTSunadjacencyMonochromeRelation, \GTSunadjMonoRel, \GTSunadjMonRel, \GTSuMR This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

```
332 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{\mathrel{\GTSuM/}}
333 \GTS@newcommand{\GTSunadjMonoRel}{\mathrel{\GTSuM/}}
334 \GTS@newcommand{\GTSunadjMonRel}{\mathrel{\GTSuM/}}
335 \GTS@newcommand{\GTSuMR}{\mathrel{\GTSuM/}}
```

\GTSdirectedAdjacencyUpMonochrome, \GTSdirAdjUpMono, \GTSdirAdjUpMon, \GTSdAUM This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```
336 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
337 \GTS@a{black}{black}{1}{}}
338 }
339 \GTS@newcommand{\GTSdirAdjUpMono}{\GTS@a{black}{black}{1}{}}
340 \GTS@newcommand{\GTSdirAdjUpMon}{\GTS@a{black}{black}{1}{}}
341 \GTS@newcommand{\GTSdAUM}{\GTS@a{black}{black}{1}{}}
```

\GTSdirectedAdjacencyUpMonochromeRelation, \GTSdirAdjUpMonoRel, \GTSdirAdjUpMonRel, \GTSdAUMR This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

```
342 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
343 \mathrel{\GTSdAUM/}}
344 }
345 \GTS@newcommand{\GTSdirAdjUpMonoRel}{\mathrel{\GTSdAUM/}}
346 \GTS@newcommand{\GTSdirAdjUpMonRel}{\mathrel{\GTSdAUM/}}
347 \GTS@newcommand{\GTSdAUMR}{\mathrel{\GTSdAUM/}}
```

\GTSnegatedDirectedAdjacencyUpMonochrome, \GTSnegDirAdjUpMono, \GTSnegDirAdjUpMon, \GTSnDAUM This macro will draw a monochrome negated directed adjacency upward symbol.

```
348 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%
349 \GTS@a{black}{black}{1}{1}{}}
350 }
351 \GTS@newcommand{\GTSnegDirAdjUpMono}{\GTS@a{black}{black}{1}{1}{}}
352 \GTS@newcommand{\GTSnegDirAdjUpMon}{\GTS@a{black}{black}{1}{1}{}}
353 \GTS@newcommand{\GTSnDAUM}{\GTS@a{black}{black}{1}{1}{}}
```

\GTSnegatedDirectedAdjacencyUpMonochromeRelation, \GTSnegDirAdjUpMonoRel, \GTSnegDirAdjUpMonRel, \GTSnDAUMR This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

```
354 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%
355 \mathrel{\GTSnDAUM/}}
356 }
357 \GTS@newcommand{\GTSnegDirAdjUpMonoRel}{\mathrel{\GTSnDAUM/}}
358 \GTS@newcommand{\GTSnegDirAdjUpMonRel}{\mathrel{\GTSnDAUM/}}
359 \GTS@newcommand{\GTSnDAUMR}{\mathrel{\GTSnDAUM/}}
```

\GTSdirectedAdjacencyDownMonochrome, \GTSdirAdjDownMono, \GTSdirAdjDownMon, \GTSdADM This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

360 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%
361 \GTS@a{black}{black}{1}{1}%
362 }
363 \GTS@newcommand{\GTSdirAdjDownMono}{\GTS@a{black}{black}{1}{1}}
364 \GTS@newcommand{\GTSdirAdjDownMonRel}{\GTS@a{black}{black}{1}{1}}
365 \GTS@newcommand{\GTSdADM}{\GTS@a{black}{black}{1}{1}}

```

\GTSdirectedAdjacencyDownMonochromeRelation, \GTSdirAdjDownMonoRel, \GTSdirAdjDownMonRel, \GTSdADMR This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

```

366 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%
367 \mathrel{\GTSdADM}/}%
368 }
369 \GTS@newcommand{\GTSdirAdjDownMonoRel}{\mathrel{\GTSdADM/}}
370 \GTS@newcommand{\GTSdirAdjDownMonRel}{\mathrel{\GTSdADM/}}
371 \GTS@newcommand{\GTSdADMR}{\mathrel{\GTSdADM/}}

```

\GTSnegatedDirectedAdjacencyDownMonochrome, \GTSnegDirAdjDownMono, \GTSnegDirAdjDownMon, \GTSnDADM This macro will draw a monochrome negated directed adjacency downward symbol.

```

372 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%
373 \GTS@a{black}{black}{1}{1}%
374 }
375 \GTS@newcommand{\GTSnegDirAdjDownMono}{\GTS@a{black}{black}{1}{1}}
376 \GTS@newcommand{\GTSnegDirAdjDownMon}{\GTS@a{black}{black}{1}{1}}
377 \GTS@newcommand{\GTSnDADM}{\GTS@a{black}{black}{1}{1}}

```

\GTSnegatedDirectedAdjacencyDownMonochromeRelation, \GTSnegDirAdjDownMonoRel, \GTSnegDirAdjDownMonRel, \GTSnDADMR This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

```

378 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%
379 \mathrel{\GTSnDADM}/}%
380 }
381 \GTS@newcommand{\GTSnegDirAdjDownMonoRel}{\mathrel{\GTSnDADM/}}
382 \GTS@newcommand{\GTSnegDirAdjDownMonRel}{\mathrel{\GTSnDADM/}}
383 \GTS@newcommand{\GTSnDADMR}{\mathrel{\GTSnDADM/}}

```

\GTSincidenceMonochrome, \GTSincMono, \GTSincMon, \GTSiM This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

384 \GTS@newcommand{\GTSincidenceMonochrome}{\GTS@i{black}{black}{1}{1}}
385 \GTS@newcommand{\GTSincMono}{\GTS@i{black}{black}{1}{1}}
386 \GTS@newcommand{\GTSincMon}{\GTS@i{black}{black}{1}{1}}
387 \GTS@newcommand{\GTSiM}{\GTS@i{black}{black}{1}{1}}

```

\GTSincidenceMonochromeRelation, \GTSincMonoRel, \GTSincMonRel, \GTSiMR This macro will draw a monochrome incidence symbol with relation spacing in math mode.

```

388 \GTS@newcommand{\GTSincidenceMonochromeRelation}{\mathrel{\GTSiM/}}
389 \GTS@newcommand{\GTSincMonoRel}{\mathrel{\GTSiM/}}
390 \GTS@newcommand{\GTSincMonRel}{\mathrel{\GTSiM/}}
391 \GTS@newcommand{\GTSiMR}{\mathrel{\GTSiM/}}

```

\GTSnegatedIncidenceMonochrome, \GTSnegIncMono, \GTSnegIncMon, \GTSnIM This macro will draw a monochrome negated incidence symbol.

```

392 \GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%
393 \GTS@i{black}{black}{1}{1}%
394 }
395 \GTS@newcommand{\GTSnegIncMono}{\GTS@i{black}{black}{1}{1}}
396 \GTS@newcommand{\GTSnegIncMon}{\GTS@i{black}{black}{1}{1}}
397 \GTS@newcommand{\GTSnIM}{\GTS@i{black}{black}{1}{1}}

```

\GTSnegatedIncidenceMonochromeRelation, \GTSnegIncMonoRel, \GTSnegIncMonRel, \GTSnIMR This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

```
398 \GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{\mathrel{\GTSnIM/}}
399 \GTS@newcommand{\GTSnegIncMonoRel}{\mathrel{\GTSnIM/}}
400 \GTS@newcommand{\GTSnegIncMonRel}{\mathrel{\GTSnIM/}}
401 \GTS@newcommand{\GTSnIMR}{\mathrel{\GTSnIM/}}
```

\GTSoutIncidenceMonochrome, \GTSoutIncMono, \GTSoutIncMon, \GTSoIM This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
402 \GTS@newcommand{\GTSoutIncidenceMonochrome}{\GTS@i{black}{black}{}}{1}{}}
403 \GTS@newcommand{\GTSoutIncMono}{\GTS@i{black}{black}{}}{1}{}}
404 \GTS@newcommand{\GTSoutIncMon}{\GTS@i{black}{black}{}}{1}{}}
405 \GTS@newcommand{\GTSoIM}{\GTS@i{black}{black}{}}{1}{}}
```

\GTSoutIncidenceMonochromeRelation, \GTSoutIncMonoRel, \GTSoutIncMonRel, \GTSoIMR This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
406 \GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{\mathrel{\GTSoIM/}}
407 \GTS@newcommand{\GTSoutIncMonoRel}{\mathrel{\GTSoIM/}}
408 \GTS@newcommand{\GTSoutIncMonRel}{\mathrel{\GTSoIM/}}
409 \GTS@newcommand{\GTSoIMR}{\mathrel{\GTSoIM/}}
```

\GTSnegatedOutIncidenceMonochrome, \GTSnegOutIncMono, \GTSnegOutIncMon, \GTSnOIM This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

```
410 \GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%
411 \GTS@i{black}{black}{1}{1}}{}}
412 }
413 \GTS@newcommand{\GTSnegOutIncMono}{\GTS@i{black}{black}{1}{1}}{}}
414 \GTS@newcommand{\GTSnegOutIncMon}{\GTS@i{black}{black}{1}{1}}{}}
415 \GTS@newcommand{\GTSnOIM}{\GTS@i{black}{black}{1}{1}}{}}
```

\GTSnegatedOutIncidenceMonochromeRelation, \GTSnegOutIncMonoRel, \GTSnegOutIncMonRel, \GTSnOIMR This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
416 \GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%
417 \mathrel{\GTSnOIM/}}{}}
418 }
419 \GTS@newcommand{\GTSnegOutIncMonoRel}{\mathrel{\GTSnOIM/}}
420 \GTS@newcommand{\GTSnegOutIncMonRel}{\mathrel{\GTSnOIM/}}
421 \GTS@newcommand{\GTSnOIMR}{\mathrel{\GTSnOIM/}}
```

\GTSinIncidenceMonochrome, \GTSinIncMono, \GTSinIncMon, \GTSiIM This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
422 \GTS@newcommand{\GTSinIncidenceMonochrome}{\GTS@i{black}{black}{}}{1}}
423 \GTS@newcommand{\GTSinIncMono}{\GTS@i{black}{black}{}}{1}}
424 \GTS@newcommand{\GTSinIncMon}{\GTS@i{black}{black}{}}{1}}
425 \GTS@newcommand{\GTSiIM}{\GTS@i{black}{black}{}}{1}}
```

\GTSinIncidenceMonochromeRelation, \GTSinIncMonoRel, \GTSinIncMonRel, \GTSiIMR This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
426 \GTS@newcommand{\GTSinIncidenceMonochromeRelation}{\mathrel{\GTSiIM/}}
427 \GTS@newcommand{\GTSinIncMonoRel}{\mathrel{\GTSiIM/}}
428 \GTS@newcommand{\GTSinIncMonRel}{\mathrel{\GTSiIM/}}
429 \GTS@newcommand{\GTSiIMR}{\mathrel{\GTSiIM/}}
```

\GTSnegatedInIncidenceMonochrome, \GTSnegInIncMono, \GTSnegInIncMon, \GTSnIIM This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

```
430 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
431 \GTS@i{black}{black}{1}{1}}%
432 }
433 \GTS@newcommand{\GTSnegInIncMono}{\GTS@i{black}{black}{1}{1}}
434 \GTS@newcommand{\GTSnegInIncMon}{\GTS@i{black}{black}{1}{1}}
435 \GTS@newcommand{\GTSnIIM}{\GTS@i{black}{black}{1}{1}}
```

\GTSnegatedInIncidenceMonochromeRelation, \GTSnegInIncMonoRel, \GTSnegInIncMon-Rel, \GTSnIIMR This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
436 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
437 \mathrel{\GTSnIIM/}%
438 }
439 \GTS@newcommand{\GTSnegInIncMonoRel}{\mathrel{\GTSnIIM/}}
440 \GTS@newcommand{\GTSnegInIncMonRel}{\mathrel{\GTSnIIM/}}
441 \GTS@newcommand{\GTSnIIMR}{\mathrel{\GTSnIIM/}}
```

\GTSmetaNot, \GTSmN This macro will draw a meta-not or relation-not symbol.

```
442 \GTS@newprotectedcommand{\GTSmN}{%
443 \text{\ensuremath{%
444 \raisebox{-0.44ex}{\scalebox{1}[1]{%
445 \raisebox{0ex}{\scalebox{1}[1]{\lnot$}}}%
446 \hspace{-1.46ex}%
447 \raisebox{1.2ex}{\scalebox{0.7}[0.7]{m}}}%
448 \hspace{0.2ex}%
449 }}}%
450 }}%
451 }
452 \GTS@newcommand{\GTSmetaNot}{\GTSmN/}
```

\GTSmetaAnd, \GTSmA This macro will draw a meta-and or relations-and symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

```
453 \GTS@newprotectedcommand{\GTSmA}{%
454 \text{\ensuremath{%
455 \raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
456 \raisebox{0ex}{\scalebox{1}[1]{\wedge$}}}%
457 \hspace{-1.46ex}%
458 \raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}}%
459 }}}%
460 }}%
461 }
462 \GTS@newcommand{\GTSmetaAnd}{\GTSmA/}
```

\GTSmetaAndRelation, \GTSmetaAndRel, \GTSmAR This macro will draw a meta-and symbol with relation spacing in math mode.

```
463 \GTS@newcommand{\GTSmetaAndRelation}{\mathrel{\GTSmA/}}
464 \GTS@newcommand{\GTSmetaAndRel}{\mathrel{\GTSmA/}}
465 \GTS@newcommand{\GTSmAR}{\mathrel{\GTSmA/}}
```

\GTSmetaOr, \GTSmO This macro will draw a meta-or or relations-or symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures ;). We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublevee or other vee symbols have taken various meanings, whilst we want an unambiguous symbol.

```
466 \GTS@newprotectedcommand{\GTSmO}{%
```

```

467 \text{\ensuremath{%
468 \raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
469 \raisebox{0ex}{\scalebox{1}[1]{\vee}}}%
470 \hspace{-1.46ex}%
471 \raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}}%
472 }}%
473 }}%
474 }
475 \GTS@newcommand{\GTSmetaOr}{\GTSmO/}

```

\GTSmetaOrRelation, \GTSmetaOrRel, \GTSmOR This macro will draw a meta-or symbol with relation spacing in math mode.

```

476 \GTS@newcommand{\GTSmetaOrRelation}{\mathrel{\GTSmO/}}
477 \GTS@newcommand{\GTSmetaOrRel}{\mathrel{\GTSmO/}}
478 \GTS@newcommand{\GTSmOR}{\mathrel{\GTSmO/}}

```

\GTSxmetallogic, \GTSxml Versatile macros for meta-logic symbols that can be given options.

Argument:

[*(optionsList)*] Sets the following options if not empty:

not or n,
and or a,
or or o,
relation or rel or r.

Of course first three options are incompatible, and the fourth is compatible only with the second and third options.

```

479 \define@key{GTSxml}{not}[1]{\def\GTS@@drawMetaNot{1}}
480 \define@key{GTSxml}{n}[1]{\def\GTS@@drawMetaNot{1}}
481 \define@key{GTSxml}{and}[1]{\def\GTS@@drawMetaAnd{1}}
482 \define@key{GTSxml}{a}[1]{\def\GTS@@drawMetaAnd{1}}
483 \define@key{GTSxml}{or}[1]{\def\GTS@@drawMetaOr{1}}
484 \define@key{GTSxml}{o}[1]{\def\GTS@@drawMetaOr{1}}
485 \define@key{GTSxml}{relation}[1]{\def\GTS@@relation{1}}
486 \define@key{GTSxml}{rel}[1]{\def\GTS@@relation{1}}
487 \define@key{GTSxml}{r}[1]{\def\GTS@@relation{1}}
488
489 \newcommand{\GTSxml}[1][]{%
490 \def\GTS@@drawMetaNot{}%
491 \def\GTS@@drawMetaAnd{}%
492 \def\GTS@@drawMetaOr{}%
493 \def\GTS@@relation{}%
494 \setkeys{GTSxml}{#1}%
495 \if\GTS@@drawMetaNot1%
496 \GTSmN/%
497 \else\if\GTS@@drawMetaAnd1%
498 \if\GTS@@relation{1}%
499 \GTSmAR/%
500 \else%
501 \GTSmA/%
502 \fi%
503 \else\if\GTS@@drawMetaOr1%
504 \if\GTS@@relation{1}%
505 \GTSmOR/%
506 \else%
507 \GTSmO/%
508 \fi%
509 \else%
510 Wrong call to \textbackslash{}GTSxml%
511 \fi%
512 \fi%
513 \fi%
514 }

```

515

516 `\newcommand{\GTSxmetallogic}[1][\GTSxml[#1]]`

\GTSx One macro to rule them all. Versatile macro for all symbols that can be given options, and a symbol type mandatory switch.

Argument:

`[\langle optionsList \rangle]` Sets the following options if not empty:

- If $\{\langle symbolType \rangle\}$ below is **a**, see `\GTSxa`:
`mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,
`secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,
`monochrome` or `mono` or `mon` or `m` (equivalent to `mC=black,sC=black`),
`negated` or `not` or `n`,
`up` or `u`,
`down` or `d`,
`relation` or `rel` or `r`.
- If $\{\langle symbolType \rangle\}$ below is **i**, see `\GTSxi`:
`mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,
`secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,
`monochrome` or `mono` or `mon` or `m` (equivalent to `mC=black,sC=black`),
`negated` or `not` or `n`,
`in` or `i`,
`out` or `o`,
`relation` or `rel` or `r`.
- If $\{\langle symbolType \rangle\}$ below is **m**, see `\GTSxml`:
`not` or `n`,
`and` or `a`,
`or` or `o`,
`relation` or `rel` or `r`.

$\{\langle symbolType \rangle\}$ a mandatory switch; it can take value **a** for adjacency, **i** for incidence, **m** for meta-logic.

517 `\newcommand{\GTSx}[2][\%`

518 `\if#2a%`

519 `\GTSxa[#1]%`

520 `\else\if#2i%`

521 `\GTSxi[#1]%`

522 `\else\if#2m%`

523 `\GTSxml[#1]%`

524 `\else%`

525 `Wrong call to \textbackslash\GTSx%`

526 `\fi%`

527 `\fi%`

528 `\fi%`

529 `\}`

530 `\endinput`

531 `\</package>`

4 Acknowledgments

Thank You God! Thank You Father! Thank You Jesus! Thank You Holy-Spirit!

5 FAQ

Question: Why `GraphTheorySymbol` instead of `GraphTheorySymbols`? `GraphTheorySymbol` does contain many symbols.

Answer: Because I followed one common usage on CTAN.

Here is the list of all current packages with the word "symbol" in their name:

actuarialsymbol, adfsymbols, compact-symbols, cookingsymbols, FdSymbol, jpnedumathsymbols, KKsymbols, listofsymbols, losymbol, maths-symbols, mdsymbol, MnSymbol, open-everyday-symbols, qsymbols, SVRsymbols, symbolindex, The Comprehensive LaTeX Symbol List, tikz-cookingsymbols, tikzsymbols.

7 "symbol" plus GraphTheorySymbol 8, 12 "symbols".

"symb" abbreviation, "sym", "sy" and "s" are also quite common;

oPlotSymb did it its own way ;) XD.

Because of the goal of GraphTheorySymbol, I was mainly looking at packages like FdSymbol, mdsymbol, MnSymbol, and The Comprehensive LaTeX Symbol List before creating my package; hence I followed this apparent majority, instead of listening to my thought that Symbols would be better.

After the package was uploaded to CTAN for a version update, this question came again, but CTAN admins were reluctant to change the package name afterward.

Otherwise, it is a matter of hours to modify and upload GraphTheorySymbols with version 2.0.0 to CTAN.

6 Support

This package is gratis and free. But if you can star it on GitHub, it is a nice encouragement.

7 Change History

v0.1.0 – 2026-08-01	v1.5.1 – 2026-09-13
General: Initial version 1	General: Use noprint option of doc package to avoid too big left margins, and use paragraph command instead 1
v1.0.0 – 2026-08-26	
General: First public release 1	
v1.1.0 – 2026-08-29	v1.5.2 – 2026-09-14
General: Added a memo 1	General: Add A4, USLetter, and Book variants of PDFs. 1
v1.1.1 – 2026-08-30	
General: Small corrections and improvements . . 1	v1.6.0 – 2026-09-18
v1.2.0 – 2026-09-06	General: Add verbatim code for compound adjacency in memo. 1
General: CTAN URL, new macros for same set of symbols 1	v1.6.1 – 2026-09-19
v1.2.1 – 2026-09-08	General: Rename test(.tex, etc.) to graph-theory-symbol-test(.tex, etc.). 1
General: Smaller tests output 1	v1.7.0 – 2026-09-21
v1.3.0 – 2026-09-08	General: Add FAQ section. Minors spacing and newpage corrections. Put table of contents and Introduction on their own page. Use tcolorbox for more visible examples.
General: New macros for monochrome symbols . 1	Disclaimer for documentation of macros with an at sign. 1
v1.4.0 – 2026-09-12	
General: New macros for symbols for logical operations on relations 1	
v1.5.0 – 2026-09-12	
General: One macro to rule them all 1	

Index

G

\GTS@@drawDownArrow	120, 121, 131, 135, 263, 264, 274, 278
\GTS@@drawMetaAnd	481, 482, 491, 497
\GTS@@drawMetaNot	479, 480, 490, 495
\GTS@@drawMetaOr	483, 484, 492, 503
\GTS@@drawSlash	115, 116, 117, 129, 134, 258, 259, 260, 272, 277
\GTS@@drawUpArrow	118, 119, 130, 135, 261, 262, 273, 278
\GTS@@mainColor	99, 100, 101, 102, 108, 110, 112, 114, 127, 134, 242, 243, 244, 245, 251, 253, 255, 257, 270, 277
\GTS@@relation	122, 123, 124, 132, 135, 265, 266, 267, 275, 278, 485, 486, 487, 493, 498, 504
\GTS@@secondColor	103, 104, 105, 106, 108, 110, 112, 114, 128, 134, 246, 247, 248, 249, 251, 253, 255, 257, 271, 277

\GTS@a	. 7, 34 , 94, 96, 141 , 142 , 143 , 147 , 148 , 149 , 153 , 154 , 155 , 159 , 160 , 161 , 167 , 168 , 169 , 173 , 174 , 175 , 320 , 321 , 322 , 323 , 328 , 329 , 330 , 331 , 337 , 339 , 340 , 341 , 349 , 351 , 352 , 353 , 361 , 363 , 364 , 365 , 373 , 375 , 376 , 377
\GTS@b	 7, 92 , 134
\GTS@drawDownArrow	 16
\GTS@drawSlash	 14
\GTS@drawUpArrow	 15
\GTS@i	. 11, 181 , 237 , 239 , 284 , 285 , 286 , 290 , 291 , 292 , 296 , 297 , 298 , 302 , 303 , 304 , 308 , 309 , 310 , 314 , 315 , 316 , 384 , 385 , 386 , 387 , 393 , 395 , 396 , 397 , 402 , 403 , 404 , 405 , 411 , 413 , 414 , 415 , 422 , 423 , 424 , 425 , 431 , 433 , 434 , 435
\GTS@j	 11, 235 , 277
\GTS@mainColor	 10, 26, 27, 53, 67, 73, 204
\GTS@mainColorBackup	 12, 26, 86, 229
\GTS@mainColorTemp	 25, 27
\GTS@newcommand	. 7, 18 , 141 , 142 , 143 , 144 , 145 , 146 , 147 , 148 , 149 , 150 , 151 , 152 , 153 , 154 , 155 , 156 , 157 , 158 , 159 , 160 , 161 , 162 , 165 , 166 , 167 , 168 , 169 , 170 , 171 , 172 , 173 , 174 , 175 , 176 , 179 , 180 , 284 , 285 , 286 , 287 , 288 , 289 , 290 , 291 , 292 , 293 , 294 , 295 , 296 , 297 , 298 , 299 , 300 , 301 , 302 , 303 , 304 , 305 , 306 , 307 , 308 , 309 , 310 , 311 , 312 , 313 , 314 , 315 , 316 , 317 , 318 , 319 , 320 , 321 , 322 , 323 , 324 , 325 , 326 , 327 , 328 , 329 , 330 , 331 , 332 , 333 , 334 , 335 , 336 , 339 , 340 , 341 , 342 , 345 , 346 , 347 , 348 , 351 , 352 , 353 , 354 , 357 , 358 , 359 , 360 , 363 , 364 , 365 , 366 , 369 , 370 , 371 , 372 , 375 , 376 , 377 , 378 , 381 , 382 , 383 , 384 , 385 , 386 , 387 , 388 , 389 , 390 , 391 , 392 , 395 , 396 , 397 , 398 , 399 , 400 , 401 , 402 , 403 , 404 , 405 , 406 , 407 , 408 , 409 , 410 , 413 , 414 , 415 , 416 , 419 , 420 , 421 , 422 , 423 , 424 , 425 , 426 , 427 , 428 , 429 , 430 , 433 , 434 , 435 , 436 , 439 , 440 , 441 , 452 , 462 , 463 , 464 , 465 , 475 , 476 , 477 , 478
\GTS@newprotectedcommand	 7, 21 , 442 , 453 , 466
\GTS@secondColor	 11, 31, 32, 45, 49, 57, 61, 192, 196, 200, 210, 216
\GTS@secondColorBackup	 13, 31, 89, 232
\GTS@secondColorTemp	 30, 32
\GTSa	 8, 141 , 144 , 145 , 146
\GTSadj	 8, 141
\GTSadjacency	 8, 141
\GTSadjacencyMonochrome	 15, 320
\GTSadjacencyMonochromeRelation	 15, 324
\GTSadjacencyRelation	 8, 144
\GTSadjMon	 15, 320
\GTSadjMono	 15, 320
\GTSadjMonoRel	 15, 324
\GTSadjMonRel	 15, 324
\GTSadjRel	 8, 144
\GTSaM	 15, 320 , 324 , 325 , 326 , 327
\GTSaMR	 15, 324
\GTSaR	 8, 144
\GTSdAD	 10, 167 , 170, 171, 172
\GTSdADM	 17, 360 , 367 , 369 , 370 , 371
\GTSdADMR	 18, 366
\GTSdADR	 10, 170
\GTSdAU	 9, 153 , 156, 157, 158
\GTSdAUM	 16, 336 , 343 , 345 , 346 , 347
\GTSdAUMR	 17, 342
\GTSdAUR	 9, 156
\GTSdirAdjDown	 10, 167
\GTSdirAdjDownMon	 17, 360
\GTSdirAdjDownMono	 17, 360
\GTSdirAdjDownMonoRel	 18, 366
\GTSdirAdjDownMonRel	 18, 366
\GTSdirAdjDownRel	 10, 170

\GTSdirAdjUp	9, 153
\GTSdirAdjUpMon	16, 336
\GTSdirAdjUpMono	16, 336
\GTSdirAdjUpMonoRel	17, 342
\GTSdirAdjUpMonRel	17, 342
\GTSdirAdjUpRel	9, 156
\GTSdirectedAdjacencyDown	10, 167
\GTSdirectedAdjacencyDownMonochrome	17, 360
\GTSdirectedAdjacencyDownMonochromeRelation	18, 366
\GTSdirectedAdjacencyDownRelation	10, 170
\GTSdirectedAdjacencyUp	9, 153
\GTSdirectedAdjacencyUpMonochrome	16, 336
\GTSdirectedAdjacencyUpMonochromeRelation	17, 342
\GTSdirectedAdjacencyUpRelation	9, 156
\GTSi	11, 284 , 287, 288, 289
\GTSiI	14, 308 , 311, 312, 313
\GTSiIM	21, 422 , 426, 427, 428, 429
\GTSiIMR	22, 426
\GTSiIR	14, 311
\GTSiM	19, 384 , 388, 389, 390, 391
\GTSiMR	19, 388
\GTSinc	11, 284
\GTSincidence	11, 284
\GTSincidenceMonochrome	19, 384
\GTSincidenceMonochromeRelation	19, 388
\GTSincidenceRelation	12, 287
\GTSincMon	19, 384
\GTSincMono	19, 384
\GTSincMonoRel	19, 388
\GTSincMonRel	19, 388
\GTSincRel	12, 287
\GTSinInc	14, 308
\GTSinIncidence	14, 308
\GTSinIncidenceMonochrome	21, 422
\GTSinIncidenceMonochromeRelation	22, 426
\GTSinIncidenceRelation	14, 311
\GTSinIncMon	21, 422
\GTSinIncMono	21, 422
\GTSinIncMonoRel	22, 426
\GTSinIncMonRel	22, 426
\GTSinIncRel	14, 311
\GTSiR	12, 287
\GTSmA	23, 453 , 463, 464, 465, 501
\GTSmAR	23, 463 , 499
\GTSmetaAnd	23, 453
\GTSmetaAndRel	23, 463
\GTSmetaAndRelation	23, 463
\GTSmetaNot	23, 442
\GTSmetaOr	23, 466
\GTSmetaOrRel	24, 476
\GTSmetaOrRelation	24, 476
\GTSmN	23, 442 , 496
\GTSmO	23, 466 , 476, 477, 478, 507
\GTSmOR	24, 476 , 505
\GTSnDAD	10, 173 , 177, 179, 180
\GTSnDADM	18, 372 , 379, 381, 382, 383

\GTSnDADMR	19, <u>378</u>
\GTSnDADR	11, <u>176</u>
\GTSnDAU	9, <u>159</u> , 163, 165, 166
\GTSnDAUM	17, <u>348</u> , 355, 357, 358, 359
\GTSnDAUMR	17, <u>354</u>
\GTSnDAUR	10, <u>162</u>
\GTSnegatedDirectedAdjacencyDown	10, <u>173</u>
\GTSnegatedDirectedAdjacencyDownMonochrome	18, <u>372</u>
\GTSnegatedDirectedAdjacencyDownMonochromeRelation	19, <u>378</u>
\GTSnegatedDirectedAdjacencyDownRelation	11, <u>176</u>
\GTSnegatedDirectedAdjacencyUp	9, <u>159</u>
\GTSnegatedDirectedAdjacencyUpMonochrome	17, <u>348</u>
\GTSnegatedDirectedAdjacencyUpMonochromeRelation	17, <u>354</u>
\GTSnegatedDirectedAdjacencyUpRelation	10, <u>162</u>
\GTSnegatedIncidence	12, <u>290</u>
\GTSnegatedIncidenceMonochrome	19, <u>392</u>
\GTSnegatedIncidenceMonochromeRelation	20, <u>398</u>
\GTSnegatedIncidenceRelation	12, <u>293</u>
\GTSnegatedInIncidence	14, <u>314</u>
\GTSnegatedInIncidenceMonochrome	22, <u>430</u>
\GTSnegatedInIncidenceMonochromeRelation	23, <u>436</u>
\GTSnegatedInIncidenceRelation	14, <u>317</u>
\GTSnegatedOutIncidence	13, <u>302</u>
\GTSnegatedOutIncidenceMonochrome	21, <u>410</u>
\GTSnegatedOutIncidenceMonochromeRelation	21, <u>416</u>
\GTSnegatedOutIncidenceRelation	13, <u>305</u>
\GTSnegDirAdjDown	10, <u>173</u>
\GTSnegDirAdjDownMon	18, <u>372</u>
\GTSnegDirAdjDownMono	18, <u>372</u>
\GTSnegDirAdjDownMonoRel	19, <u>378</u>
\GTSnegDirAdjDownMonRel	19, <u>378</u>
\GTSnegDirAdjDownRel	11, <u>176</u>
\GTSnegDirAdjUp	9, <u>159</u>
\GTSnegDirAdjUpMon	17, <u>348</u>
\GTSnegDirAdjUpMono	17, <u>348</u>
\GTSnegDirAdjUpMonoRel	17, <u>354</u>
\GTSnegDirAdjUpMonRel	17, <u>354</u>
\GTSnegDirAdjUpRel	10, <u>162</u>
\GTSnegInc	12, <u>290</u>
\GTSnegIncMon	19, <u>392</u>
\GTSnegIncMono	19, <u>392</u>
\GTSnegIncMonoRel	20, <u>398</u>
\GTSnegIncMonRel	20, <u>398</u>
\GTSnegIncRel	12, <u>293</u>
\GTSnegInInc	14, <u>314</u>
\GTSnegInIncMon	22, <u>430</u>
\GTSnegInIncMono	22, <u>430</u>
\GTSnegInIncMonoRel	23, <u>436</u>
\GTSnegInIncMonRel	23, <u>436</u>
\GTSnegInIncRel	14, <u>317</u>
\GTSnegOutInc	13, <u>302</u>
\GTSnegOutIncMon	21, <u>410</u>
\GTSnegOutIncMono	21, <u>410</u>
\GTSnegOutIncMonoRel	21, <u>416</u>
\GTSnegOutIncMonRel	21, <u>416</u>
\GTSnegOutIncRel	13, <u>305</u>

\GTSnI	12, 290, 293, 294, 295
\GTSnII	14, 314, 317, 318, 319
\GTSnIIM	22, 430, 437, 439, 440, 441
\GTSnIIMR	23, 436
\GTSnIIR	14, 317
\GTSnIM	19, 392, 398, 399, 400, 401
\GTSnIMR	20, 398
\GTSnIR	12, 293
\GTSnOI	13, 302, 305, 306, 307
\GTSnOIM	21, 410, 417, 419, 420, 421
\GTSnOIMR	21, 416
\GTSnOIR	13, 305
\GTSoI	13, 296, 299, 300, 301
\GTSoIM	20, 402, 406, 407, 408, 409
\GTSoIMR	21, 406
\GTSoIR	13, 299
\GTSoutInc	13, 296
\GTSoutIncidence	13, 296
\GTSoutIncidenceMonochrome	20, 402
\GTSoutIncidenceMonochromeRelation	21, 406
\GTSoutIncidenceRelation	13, 299
\GTSoutIncMon	20, 402
\GTSoutIncMono	20, 402
\GTSoutIncMonoRel	21, 406
\GTSoutIncMonRel	21, 406
\GTSoutIncRel	13, 299
\GTSsetMainColor	7, 24, 36, 86, 183, 229
\GTSsetSecondColor	7, 29, 39, 89, 186, 232
\GTSu	8, 147, 150, 151, 152
\GTSuM	15, 328, 332, 333, 334, 335
\GTSuMR	16, 332
\GTSunadj	8, 147
\GTSunadjacency	8, 147
\GTSunadjacencyMonochrome	15, 328
\GTSunadjacencyMonochromeRelation	16, 332
\GTSunadjacencyRelation	9, 150
\GTSunadjMon	15, 328
\GTSunadjMono	15, 328
\GTSunadjMonoRel	16, 332
\GTSunadjMonRel	16, 332
\GTSunadjRel	9, 150
\GTSuR	9, 150
\GTSx	24, 517
\GTSxa	8, 99, 519
\GTSxadj	8, 99
\GTSxadjacency	8, 99
\GTSxi	11, 242, 521
\GTSxinc	11, 242
\GTSxincidence	11, 242
\GTSxmetallogic	24, 479
\GTSxml	24, 479, 523